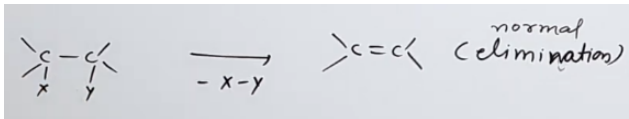
# ORGANIC REACTION MECHANISM – IV

ORM-IV

## Elimination Reactions (Reverse reaction of addition)




### On the basis of Relative position of eliminating group

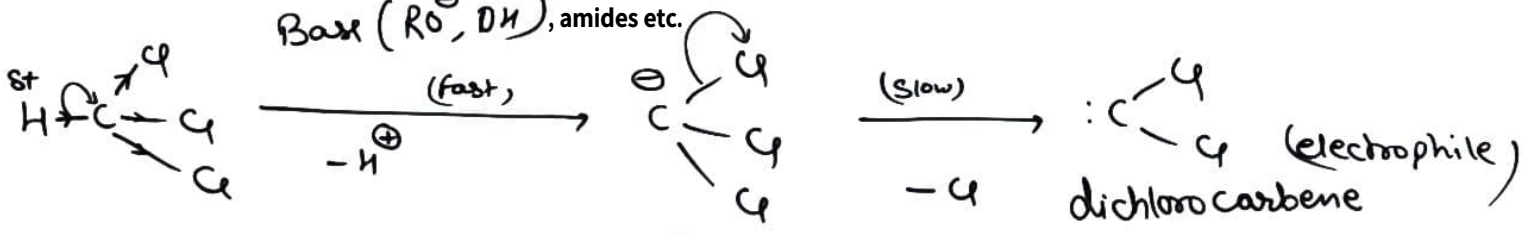
- α. elimination
- β. elimination. (Imp)
  - $E_1$ (Unimolecular elimination Rxn)
  - $E_2$ (Bimolecular " " ") (Anti)
  - $E_i$ (Internal " " ") (Syn)
  - $E_{1CB}$ (Unimolecular conjugate base elimination.
- γ. elimination

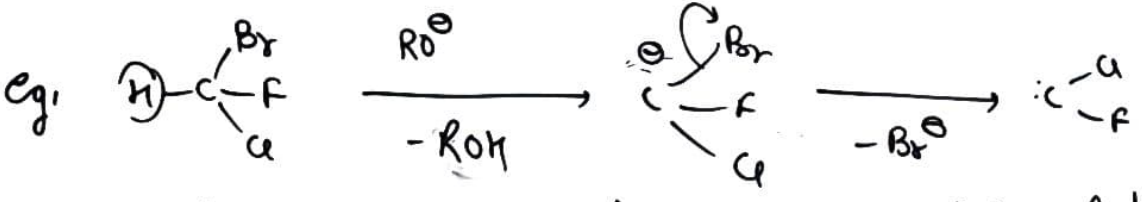
### On the basis of Stereochemistry of elimination Rxn (for stereospecific elim.)

- Syn elimination
  **when two groups remove from same plane**
- Anti elimination.
  **when two groups remove from opposite planes**

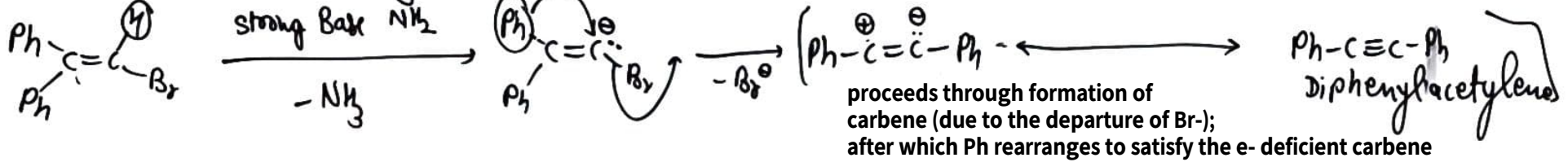
When two group are eliminate from same position.

Example 1 Formation of carbene from chloroform




Example 2 : Formation of alkyne from 2-Bromo-1,1-diphenylethene



# ORGANIC REACTION MECHANISM – IV  ORM-IV

## Elimination Reactions (Reverse reaction of addition)

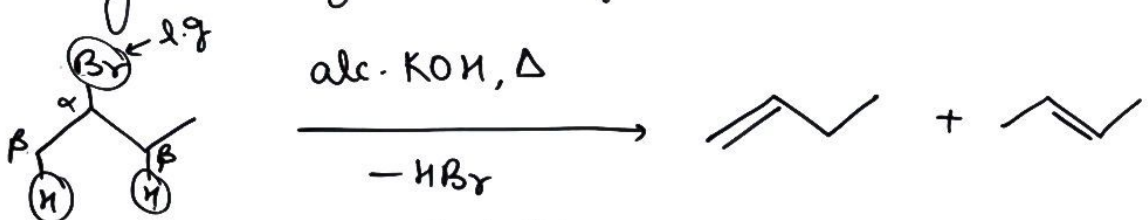
### [B] β-Elimination = (1, 2 - elimination) :-

the OG elimination reaction

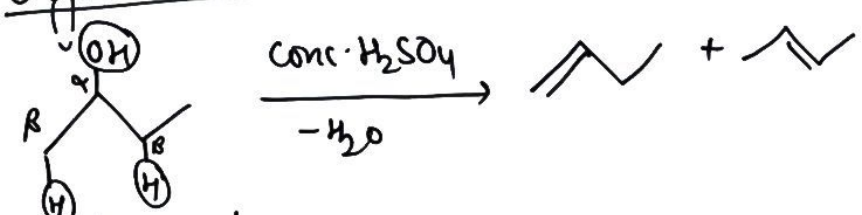
When two groups are eliminate from adjacent position ( form π bond)

Examples.

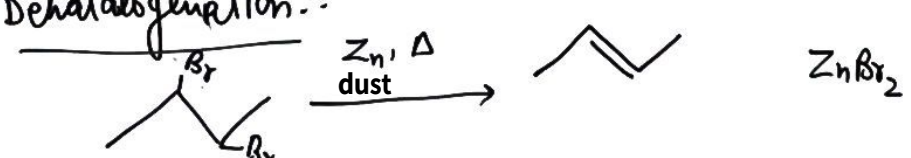
① Dehydrohalogenation of alkyl halide.




② Dehydration in alcohols:-




③ Dehalalogenation:-



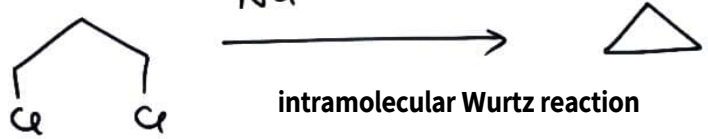
# ORGANIC REACTION MECHANISM – IV

ORM-IV

## Elimination Reactions (Reverse reaction of addition)

### [C] γ- Elimination (1-3- elimination)

When 2 groups eliminate from 1 and 3rd position. and formation of 3 membered ring takes place

eg 


intramolecular Wurtz reaction

theoreticallly, 1-4 elimination and 1-5 elimination and higher are also possible

# ORGANIC REACTION MECHANISM – IV  ORM-IV

## (I) $E_2$ (Bimolecular elimination Reaction)

concerted step: bond breaking and formation takes place in the same single step

① It is single step concerted reaction.

② $E_2$ reaction is possible in presence of strong base, Δ

for example.

heat since the rex. is endothermic

- ⇒ Alc. KOH, Δ
- ⇒ R-OH + KOH, Δ

in alcoholic KOH, the base is actually RO-, which causes elimination
aq. KOH prefers SN2 substitution over elimination

- ⇒ $R-O^{\ominus}$, Δ
- ⇒ $\overset{\ominus}{N}H_2$ Δ
- ⇒ $R^{\ominus}$ Δ etc.

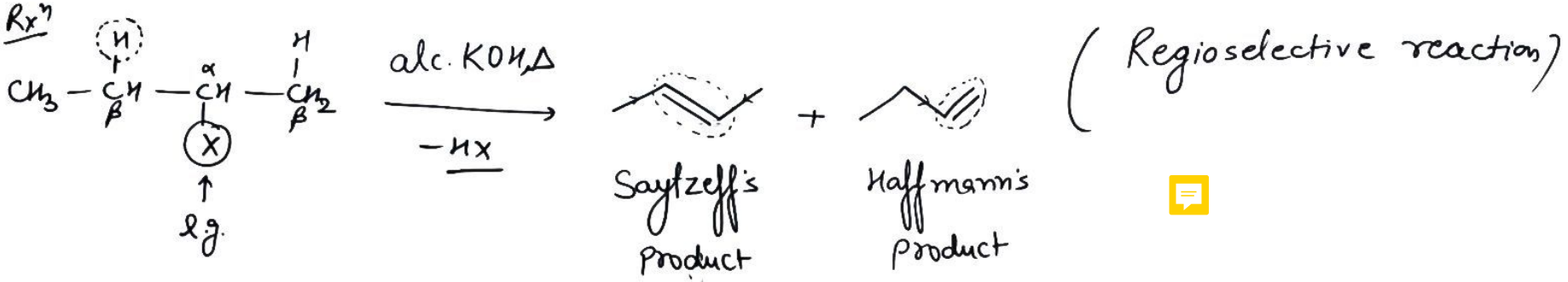
(3)

# ORGANIC REACTION MECHANISM – IV  ORM-IV

## (I) $E_2$ (Bimolecular elimination Reaction)

### (3) Reaction and mechanism.

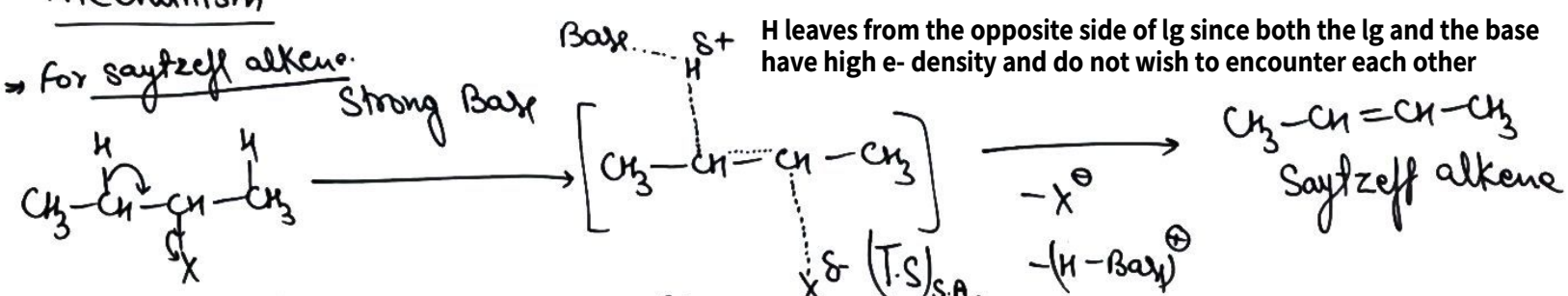
→ Rxn




( Regioselective reaction)

**mechanism**

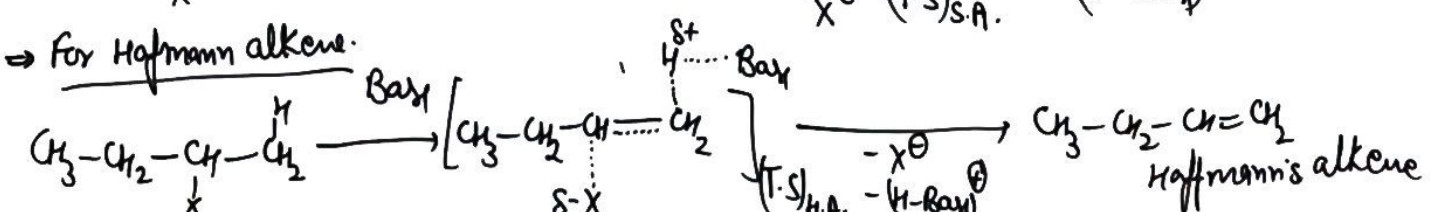
→ For saytzeff alkene.




**H leaves from the opposite side of lg since both the lg and the base have high e- density and do not wish to encounter each other**

**hence, the rex. is anti elimination**

→ For Hoffmann alkene.




**S.A = Saytzeff Alkene**
**H.A = Hoffmann Alkene**

# ORGANIC REACTION MECHANISM – IV

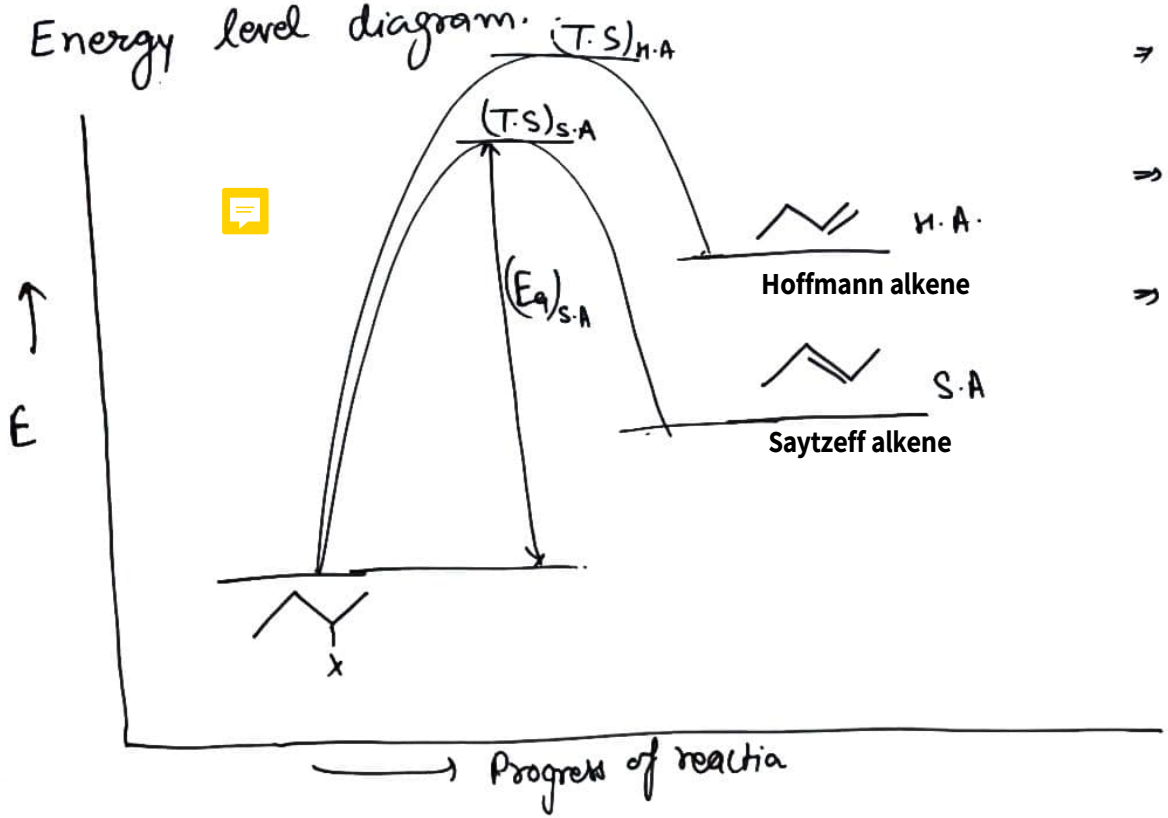
ORM-IV

## (I) $E_2$ (Bimolecular elimination Reaction)

### (3) Reaction and mechanism.

⇒ Elimination Rxⁿ are endothermic reaction. (So favairable on high temp.)

⇒ Energy level diagram.




⇒ Rate of $E_2 \propto [R-X]$ [Base]

⇒ 2nd order Rxⁿ.

⇒ Anti elimination, Stereospecific (elimination of l.g and hydrogen of β-position of l.g eliminate from opp. plane)

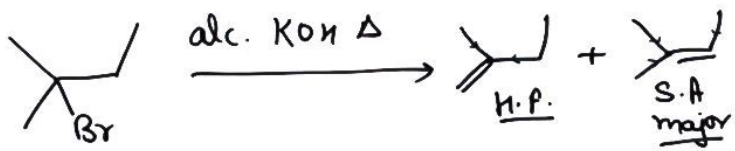
(AT, TAE)

# ORGANIC REACTION MECHANISM – IV  ORM-IV

## (I) $E_2$ (Bimolecular elimination Reaction)

### (4) Sayfzeff's Rule = Zaitsev's Rule:-

Generally more stable alkene (more alkylated alkene) forms as major product called Sayfzeff alkene.




### (5) Reactivity of $E_2$ elimination ∝ Stability of T.S.

T.S also has alkene character, so indirectly, stability of T.S depends on the stability of the alkene product

" " " " ∝ (stability of alkene product)

⇒ Reactivity of $E_2$ elim..... ∝ Stability of l.g.

($R-I > R-Br > R-Cl > R-F$ Reactivity for $E_2$)

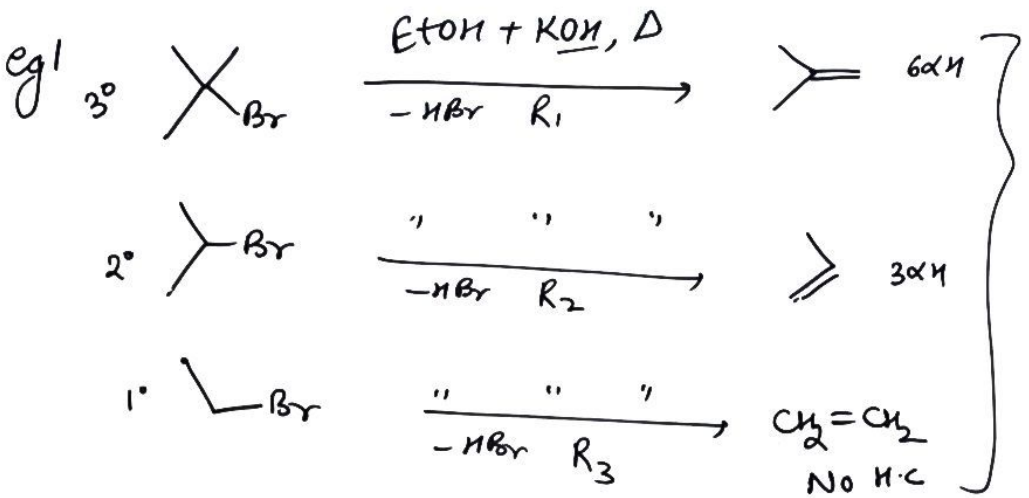
unlike substitution, R-F here shows E2 elimination

→ Reactivity on the basis of degree of alkyl halide ⇒ $3^\circ R-X > 2^\circ R-X > 1^\circ R-X$

# ORGANIC REACTION MECHANISM – IV

ORM-IV

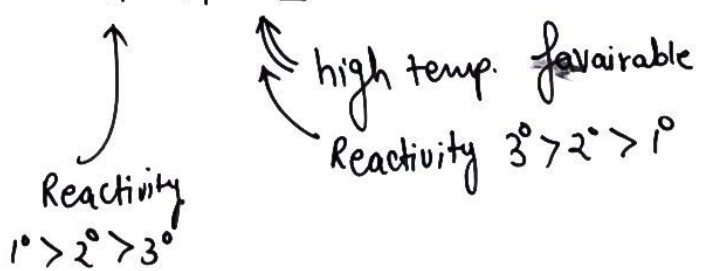
## (I) $E_2$ (Bimolecular elimination Reaction)




Reactivity = $R_1 > R_2 > R_3$

Stability of Product $R_1 > R_2 > R_3$

(6) $S_N2$ || $E_2$ — **both take place parallelly in varying amounts**




**we can also comment upon the ratio of the rates of SN2 and E2 in different species, by comparing the on the basis of degree and or temperature**

## (I) $E_2$ (Bimolecular elimination Reaction)

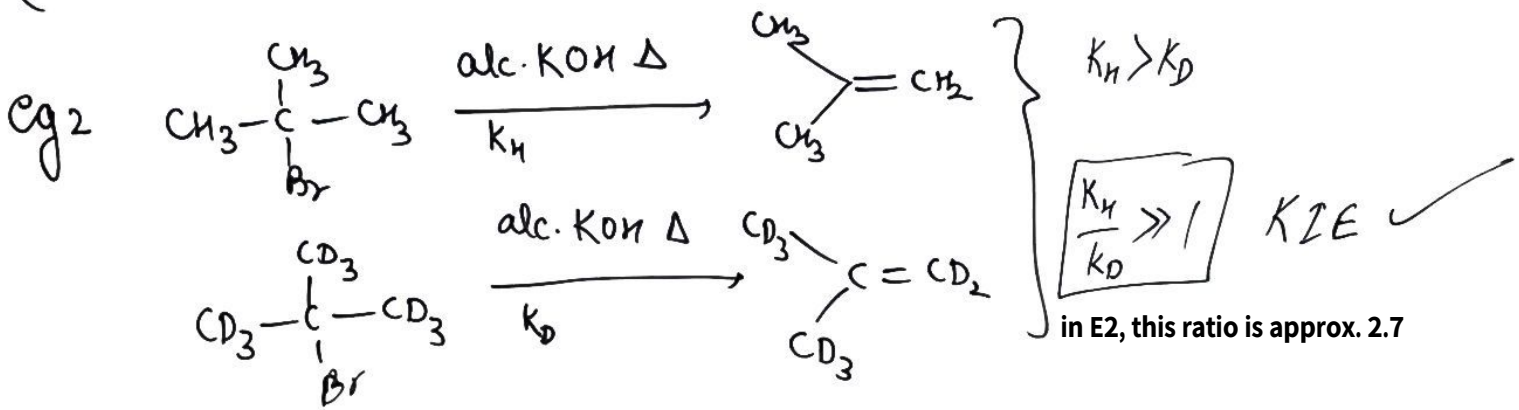
(7) Rearrangement Not possible in $E_2$

(8) Solvent = Polar aprotic solvent

as in polar protic solvents, the base will react with the acidic H

but what about alcoholic KOH? there, it is not in alcoholic form but RO- form, which is astronger base than KOH and more suitable for E2

(9) Kinetic Isotopic Effect (KIE) applicable in $E_2$ elimination (C–H Bond dissociate in RDS)

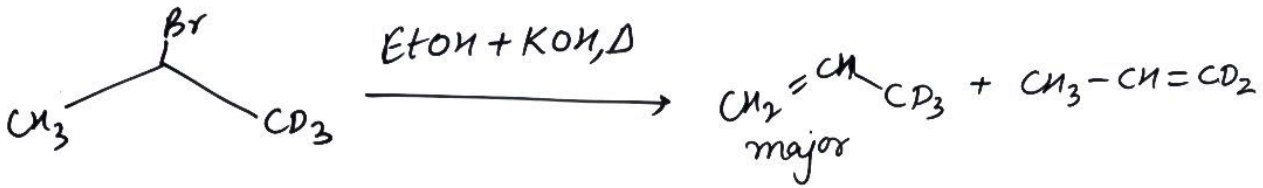


# ORGANIC REACTION MECHANISM – IV

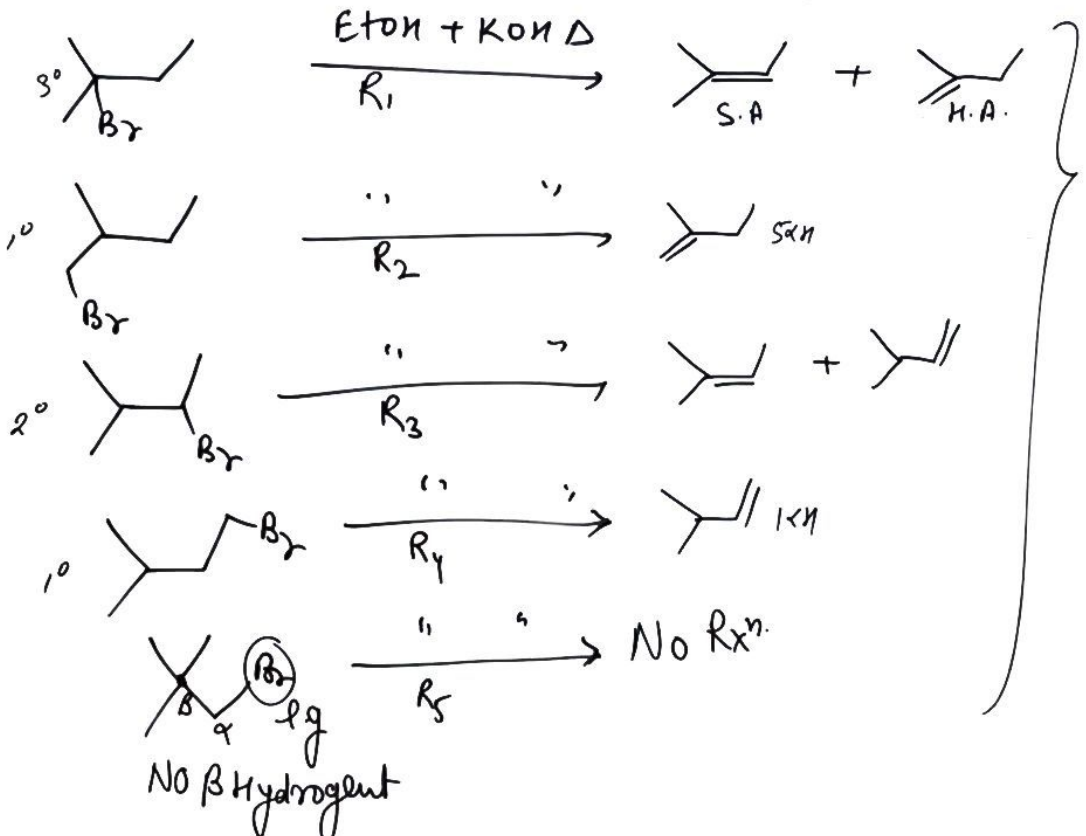


## (I) $E_2$ (Bimolecular elimination Reaction)

eg3




eg4




Reactivity

$R_1 > R_3 > R_2 > R_4 > R_5$

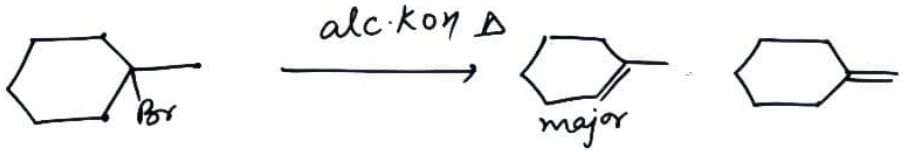
**by comparing the degree of substrates and stability of the alkene formed**

# ORGANIC REACTION MECHANISM – IV

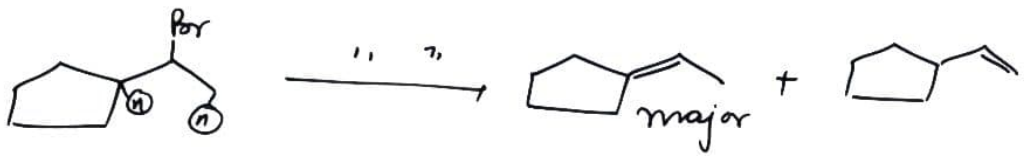


## (I) $E_2$ (Bimolecular elimination Reaction)

eg 5



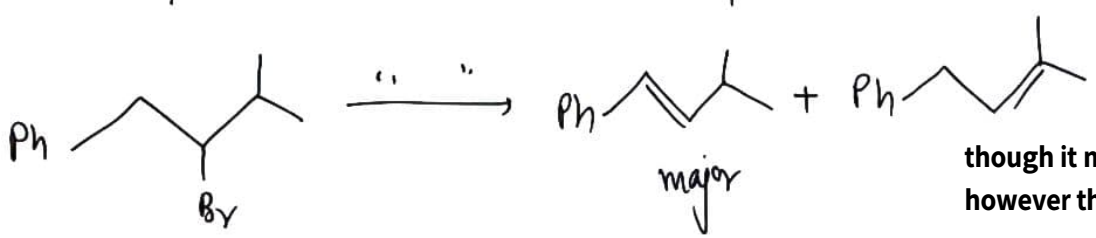

eg 6




eg 7




eg 8

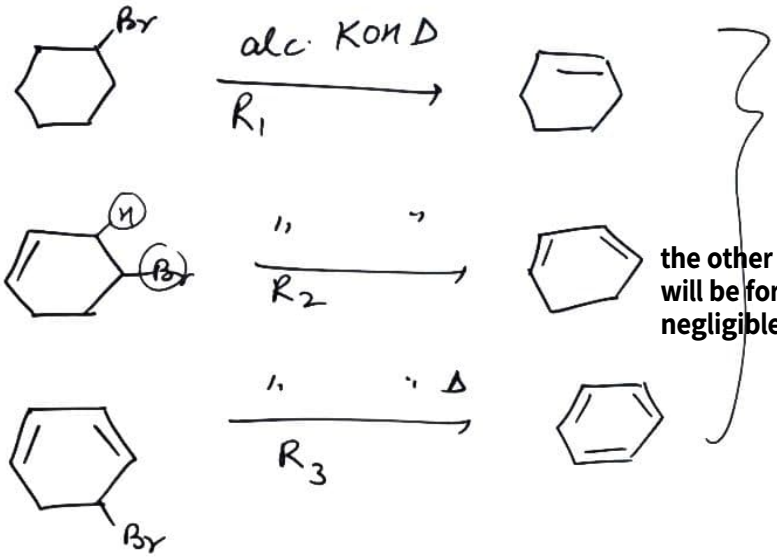



**though it more hyperconjugation,**
**however there is resonance in the first product**

# ORGANIC REACTION MECHANISM – IV

## (I) $E_2$ (Bimolecular elimination Reaction)

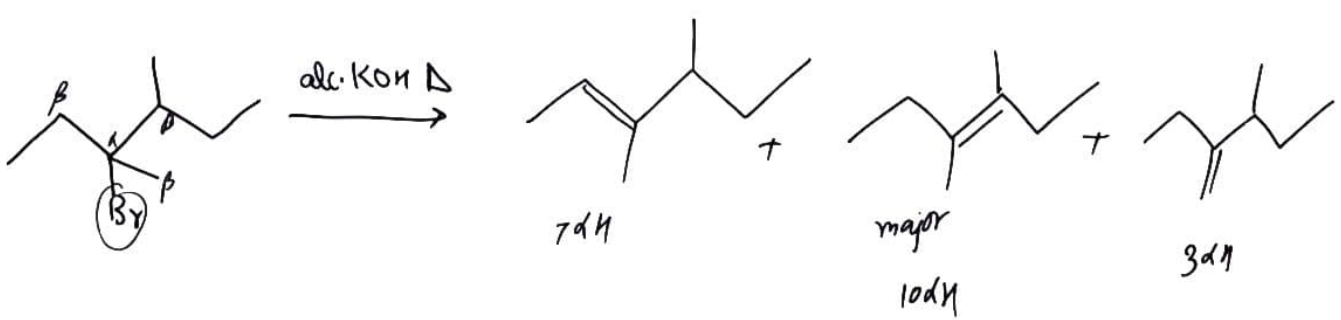
eg 9




$R_3 > R_2 > R_1$ = Reactivity.

the other product will be formed in practically negligible amount

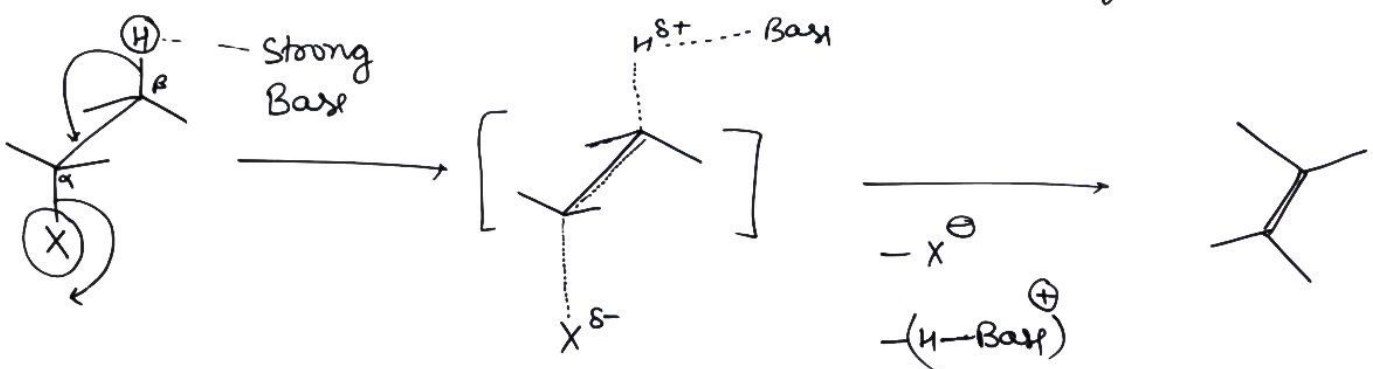
eg 10



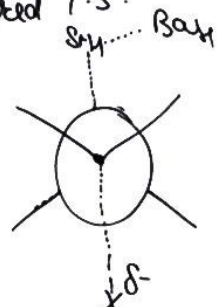
# ORGANIC REACTION MECHANISM -IV (ORM-IV)

## (I) $E_2$ Elimination :-

(9) $E_2$ elimination is stereoselective as well as stereospecific, anti elimination reaction. (via. anti periplanar T.S. = Anti conformation = Staggered conformation)




Anti periplanar Staggered T.S.



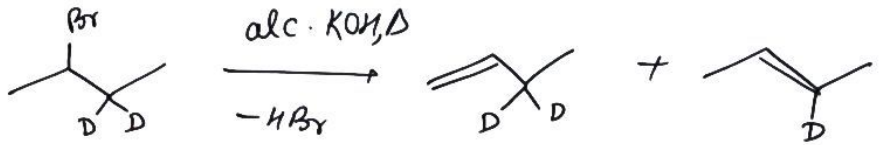

**if formation of anti-confirmation is not possible, E2 elimination will not be possible**

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# ORGANIC REACTION MECHANISM -IV (ORM-IV)

## (I) $E_2$ Elimination :-

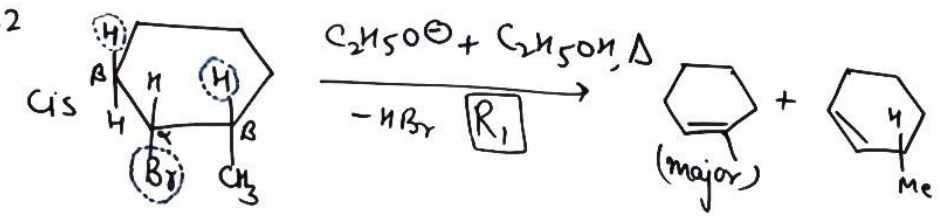
eg 11

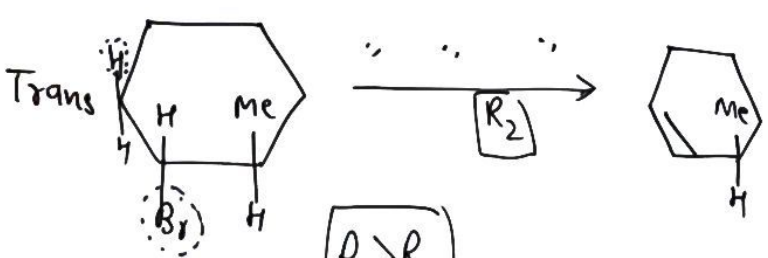



more stable alkene (major)

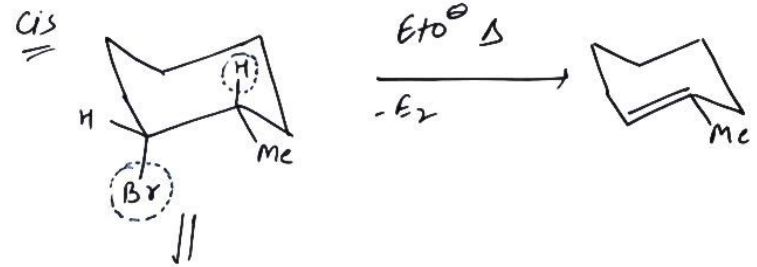
Stability of alkene > KIE

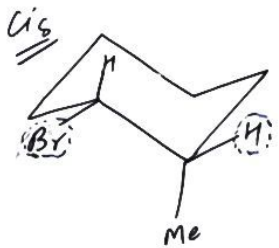
eg 12



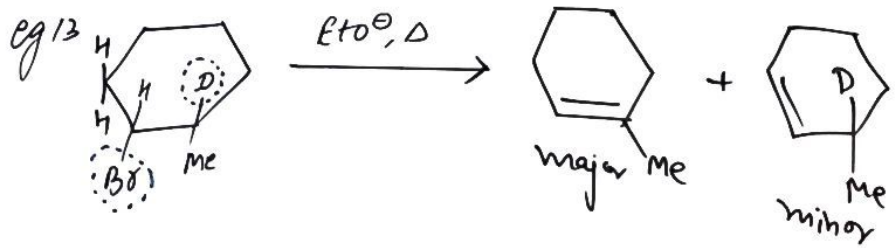

$R_1 > R_2$




**the equatorial pos. are not exactly opposite to each other**
**hence, equatorial conformation undergoes flipping to the axial conformation and E2 elimination takes place**

eg 13

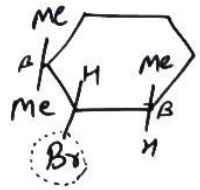



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# ORGANIC REACTION MECHANISM -IV (ORM-IV)

## (I) $E_2$ Elimination :-

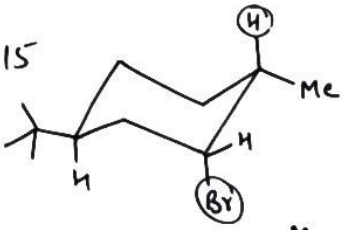
eg 14




alc. KOH Δ → No Rxn

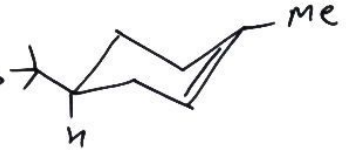
even SN2 rex. does not take place due to the crowding

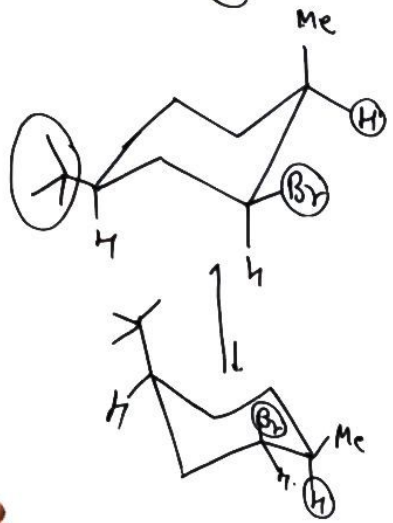
eg 15



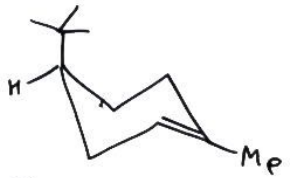

alc. KOH Δ

$R_1$ − HBr



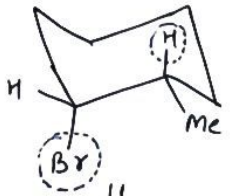

$R_2$




the tert-butyl grp. is extremely unstable in the axial position

$R_1 >>>> R_2$

cis

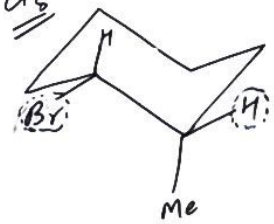



$EtO^{\ominus}$ Δ

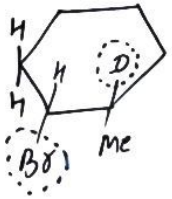
$-E_2$



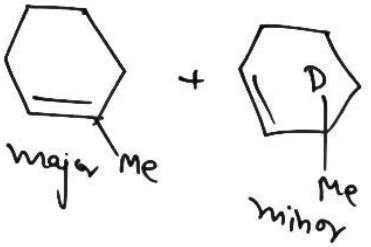

cis




eg 13




$EtO^{\ominus}$, Δ



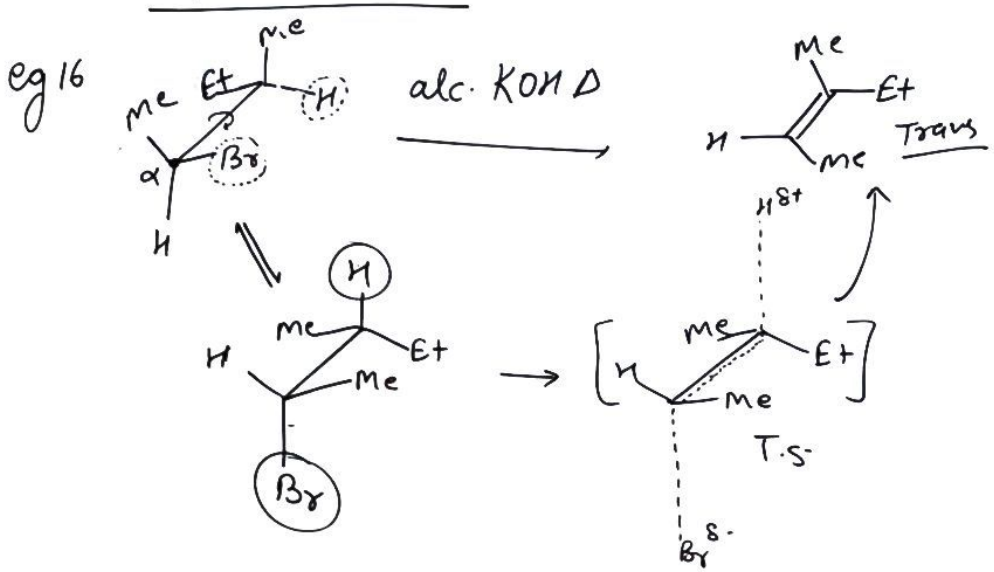

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# ORGANIC REACTION MECHANISM -IV (ORM-IV)

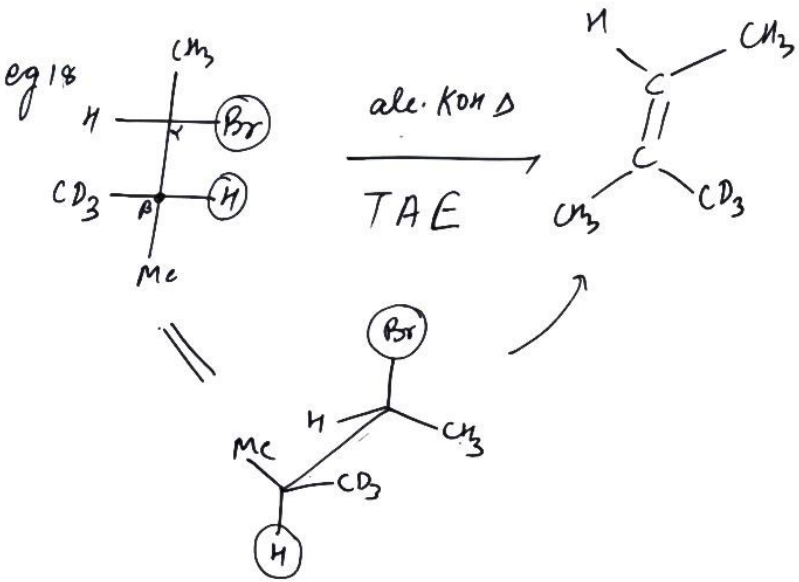
(CAT)

## (I) $E_2$ Elimination :-

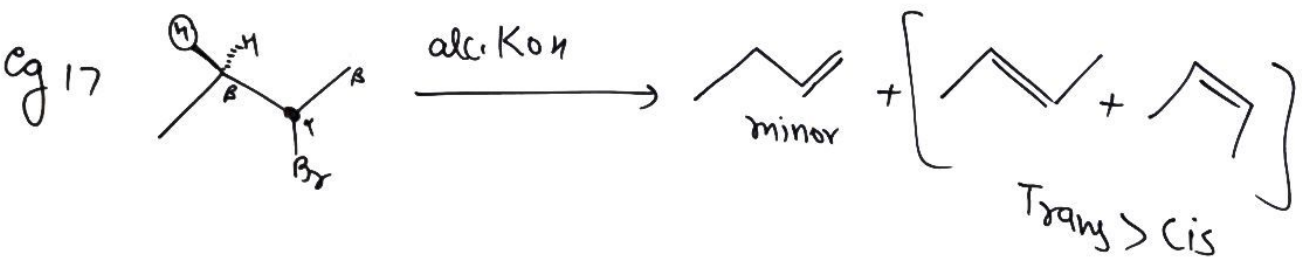
eg 16




eg 18




eg 17




the actual structure is threo

however, nikalne wale groups are erythro to each other

hamesha ye CAT/TAE wala jugaad lagate hue dhyan rkho ki judne/nikalne wale ke respect pe threo/erythro dekhna hai, actual ka nahi

cis trans bhi vertical wale group ke respect m dekhna

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# ORGANIC REACTION MECHANISM -IV (ORM-IV)

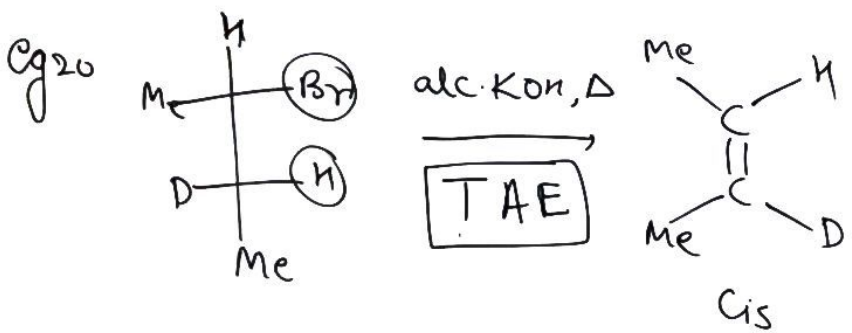
(CAT)

## (I) $E_2$ Elimination :-

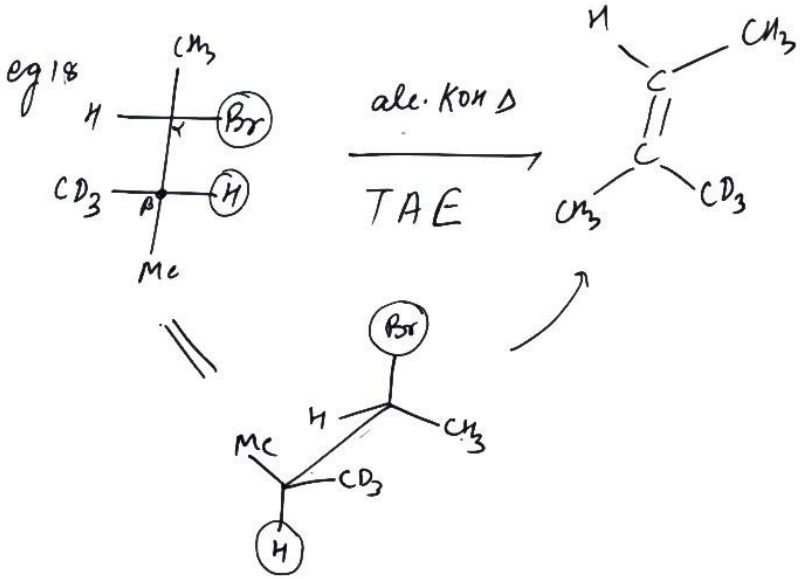
eg 19




eg 20




eg 18

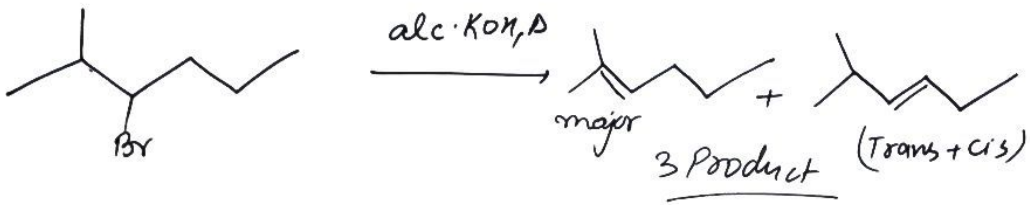



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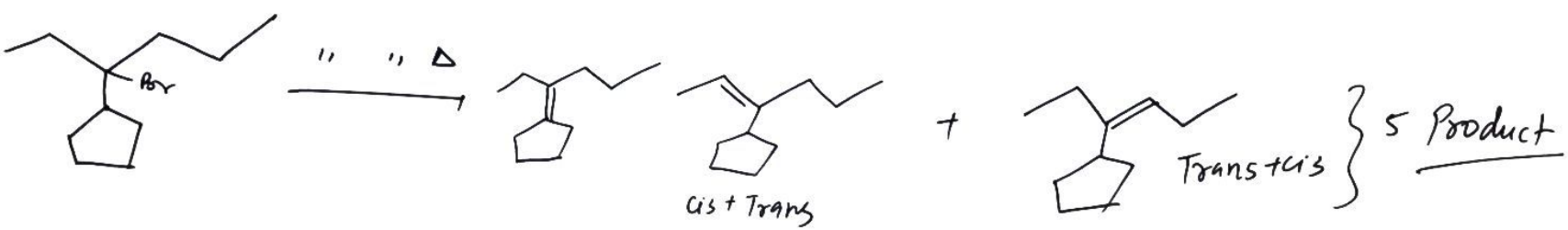
# ORGANIC REACTION MECHANISM -IV (ORM-IV)

## (I) $E_2$ Elimination :-

eg 21



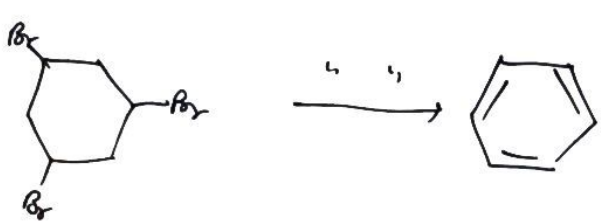

eg 22




Eg 22




Eg 23

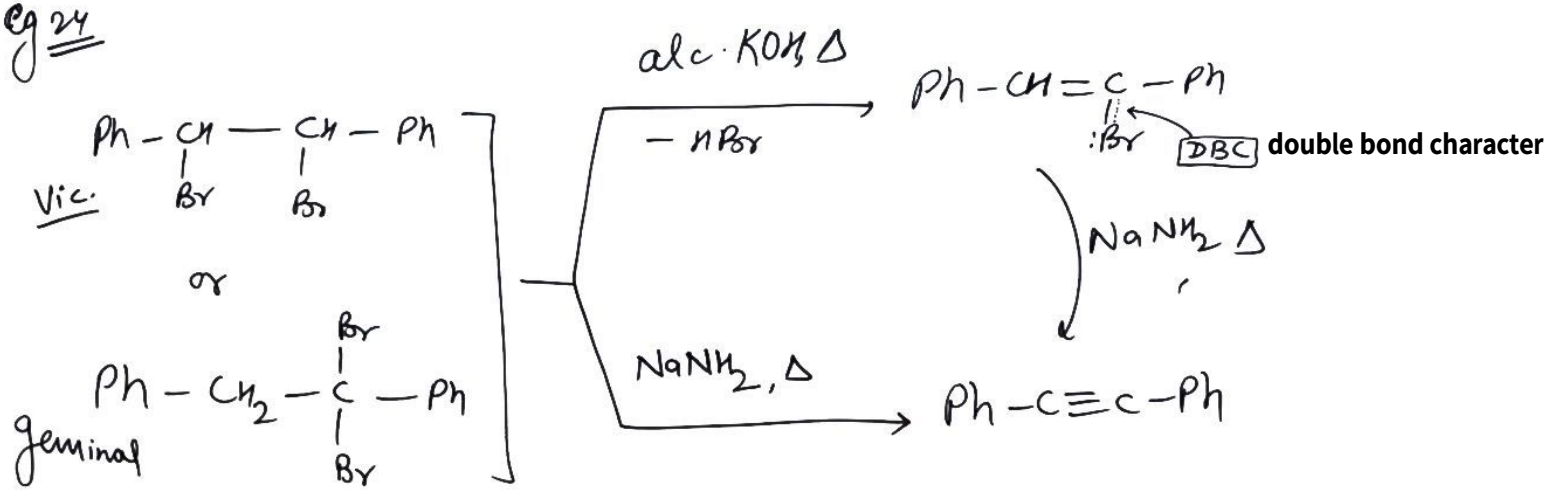



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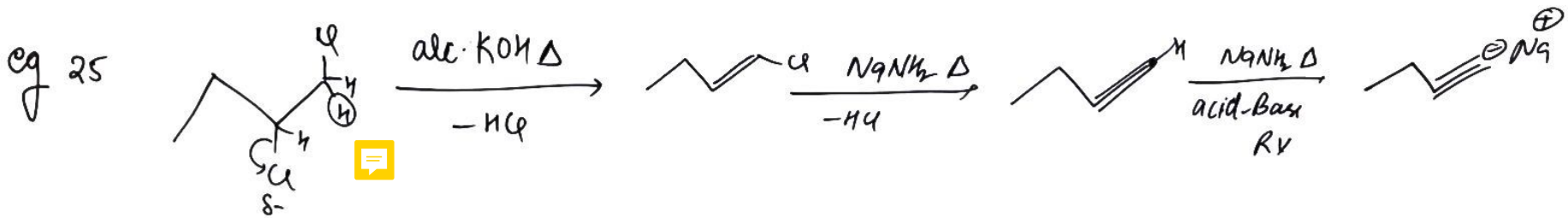
# ORGANIC REACTION MECHANISM -IV (ORM-IV)

## (I) $E_2$ Elimination :-

eg 24




eg 25



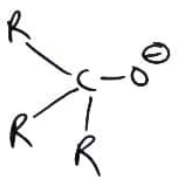

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# ORGANIC REACTION MECHANISM -IV (ORM-IV)

## (I) $E_2$ Elimination :-

**Note** :- Generally in $E_2$ elimination Saytzeff's product form as major product but in presence of following conditions Hoffmann's can form as major product.

ⓐ In presence of bulky base :- (Steric hindrance)



, $3^\circ$ amine $\Delta$

ⓑ In presence of strong $e^-$ withdrawing l.g. like ( $-F$, $-\overset{\oplus}{N}R_3$, $-\overset{\oplus}{S}R_2$ )

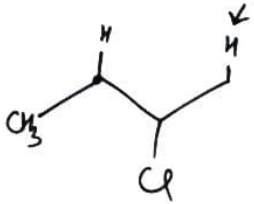
(c) In presence crowding on $\gamma$ position on l.g.

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# ORGANIC REACTION MECHANISM -IV (ORM-IV)

## (I) $E_2$ Elimination :-

eg 26




← less crowded

$C_2H_5O^{\ominus}$, Δ

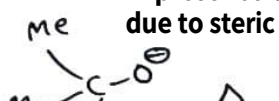
or

$C_2H_5OH + KOH$ Δ

major + 

agar alcohol likhke chod de, assume chota alcohol

in presence of bulky base, least crowded β-H will eliminate due to steric hindarance

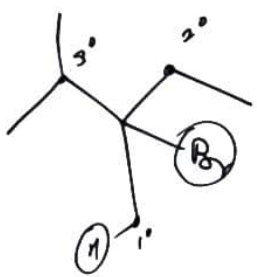



or

$t\text{-Bu-OH} + KOH$ Δ

minor + major

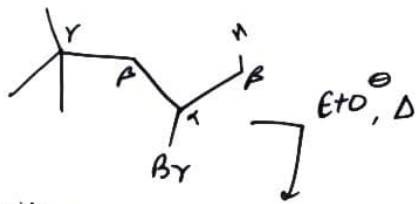
eg 27




$C_2H_5O^{\ominus}$ Δ → major

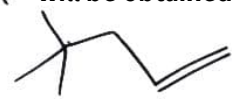
$t\text{-BuO}^{\ominus}$, Δ → major

eg 28




γ-position highly crowded

irrespective of base, Hoffmann product will be obtained in major amount



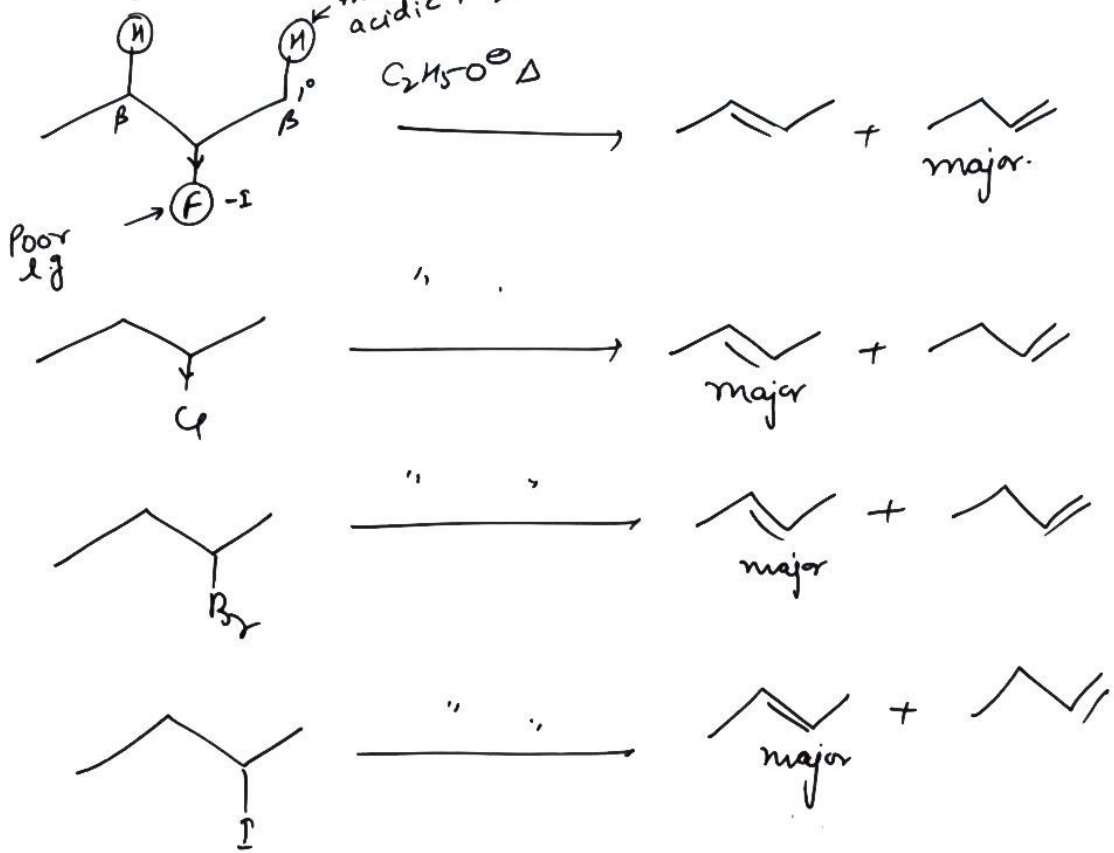
major

# ORGANIC REACTION MECHANISM -IV (ORM-IV)

## (I) $E_2$ Elimination :-

more acidic β H due to the -I effect of F

Eg 29

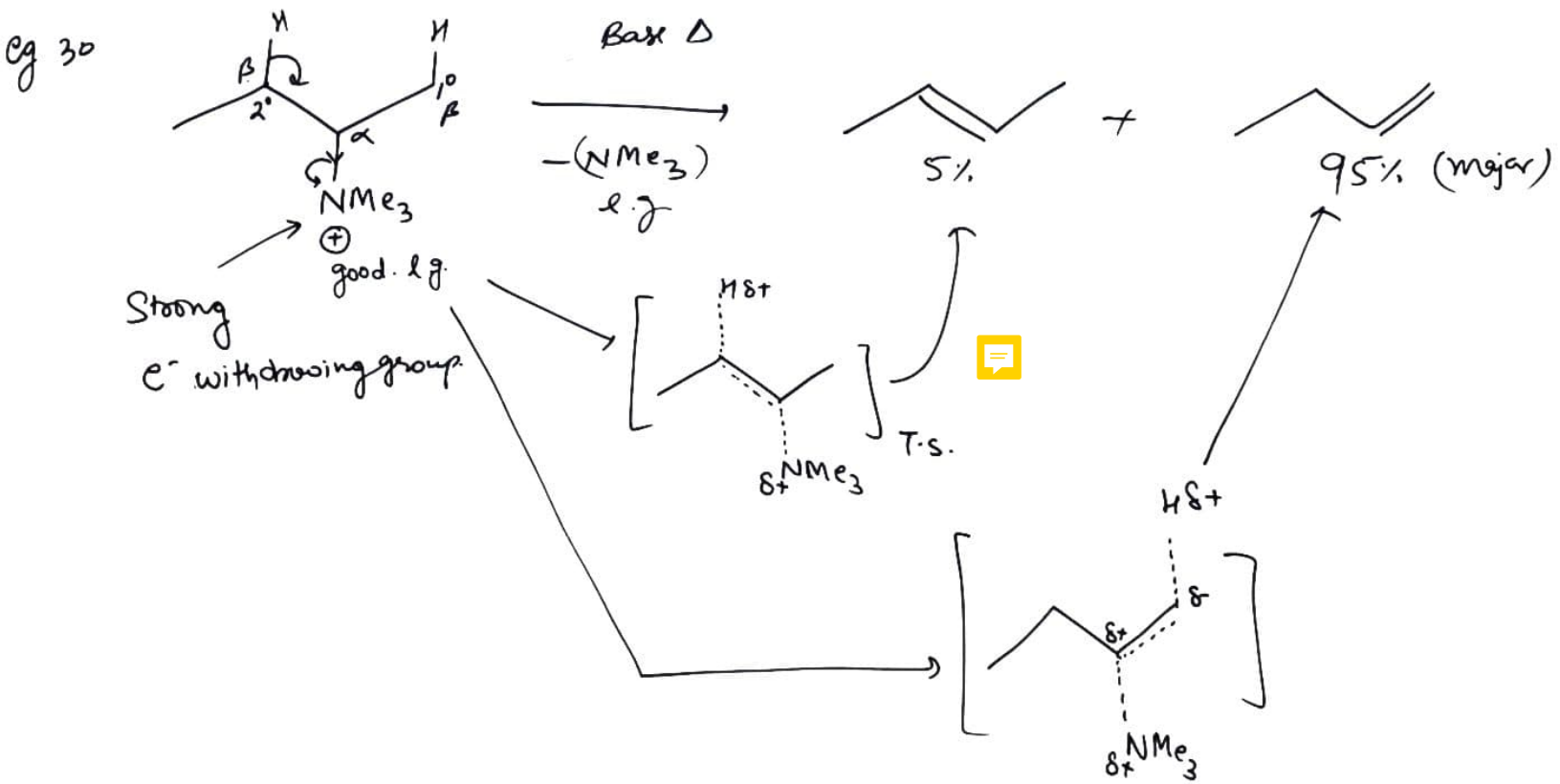



% of Hoffmann's product ↓

% of Hofmann product ∝ $\frac{1}{\text{l.g. ability}}$

# ORGANIC REACTION MECHANISM -IV (ORM-IV)

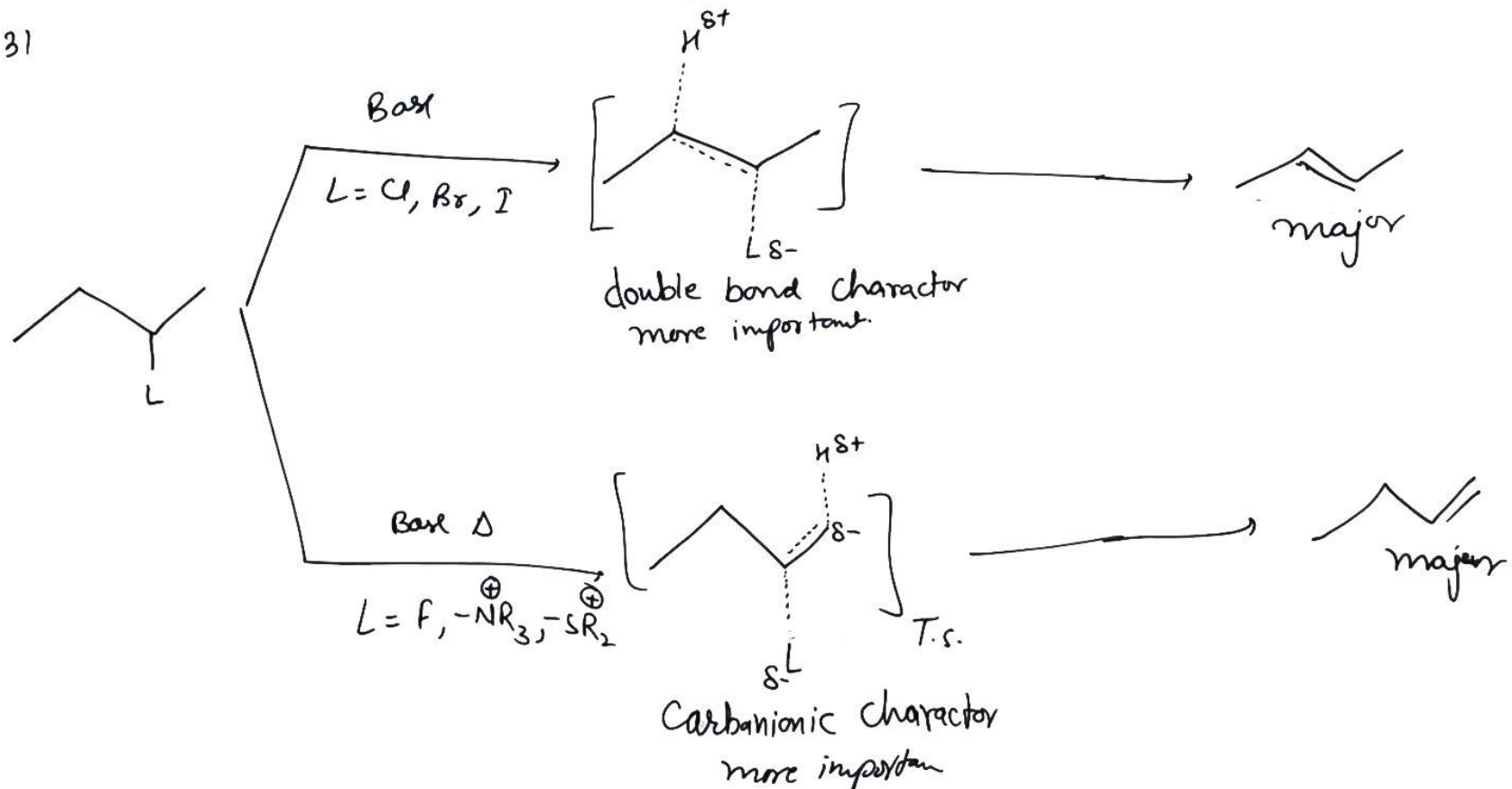
## (I) $E_2$ Elimination :-



# ORGANIC REACTION MECHANISM -IV (ORM-IV)

## (I) $E_2$ Elimination :-

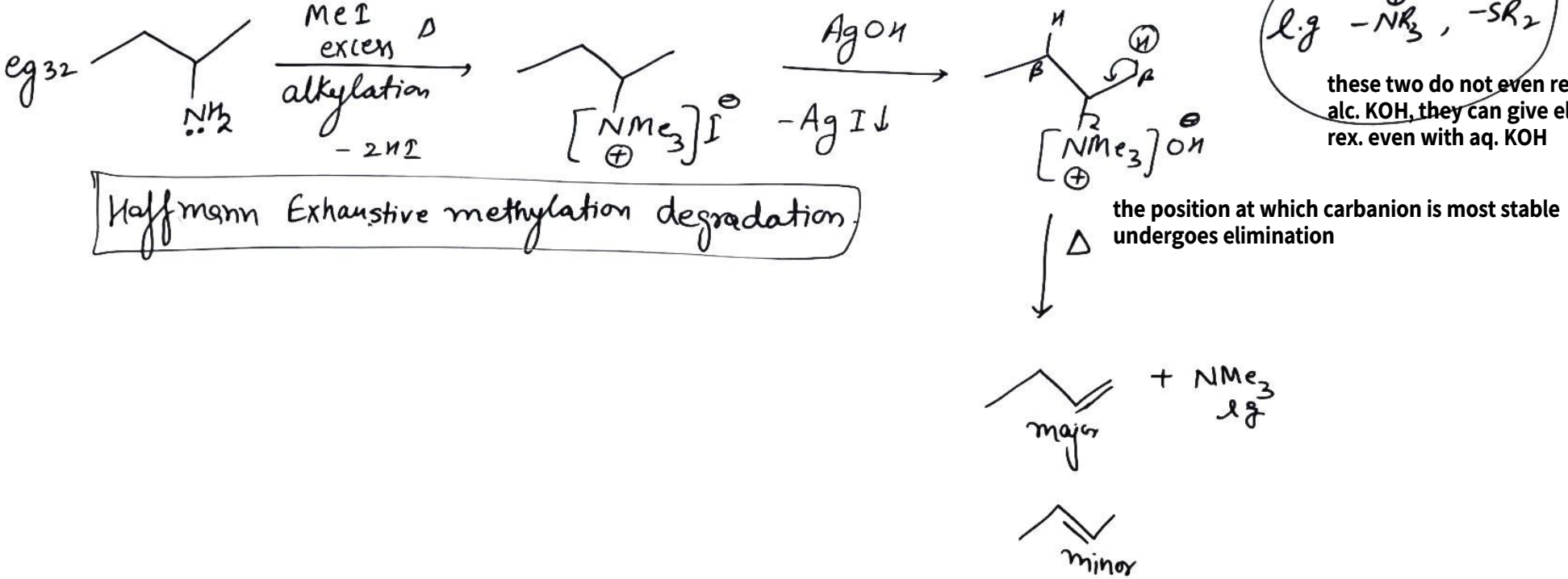
eg 31




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# ORGANIC REACTION MECHANISM -IV (ORM-IV)

## (I) $E_2$ Elimination :-



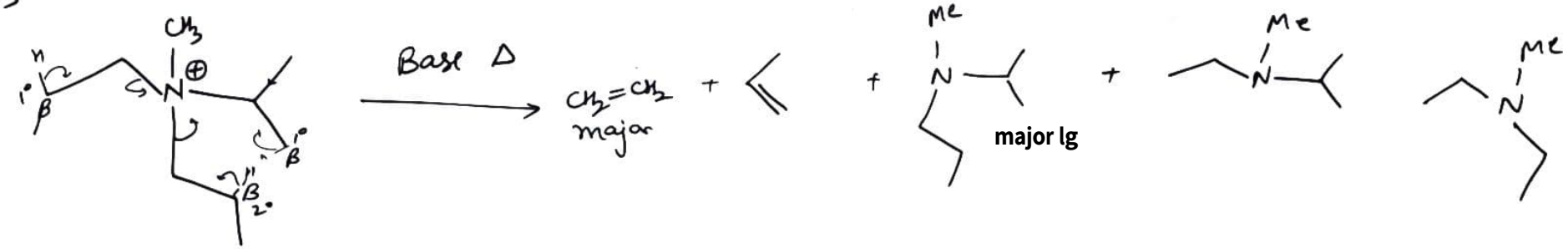

**Hoffmann Exhaustive methylation degradation**

# ORGANIC REACTION MECHANISM -IV (ORM-IV)

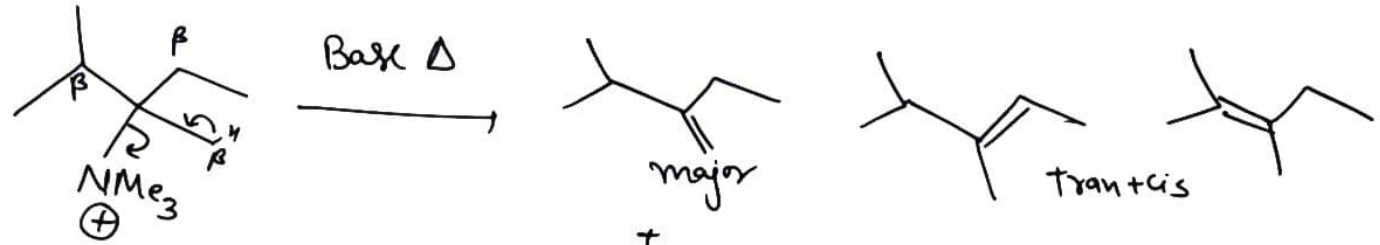
## (I) $E_2$ Elimination :-

acidic strength decreased
due to +I of Me

eg 33




eg 34




+ $NMe_3$ lg

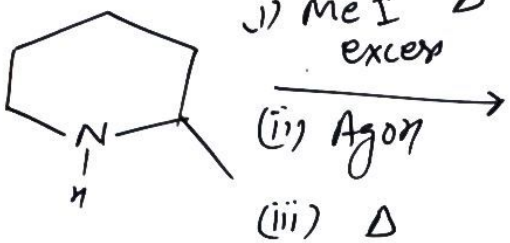


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# ORGANIC REACTION MECHANISM -IV (ORM-IV)

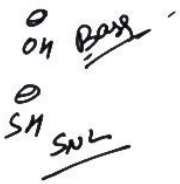
## (I) $E_2$ Elimination :-

eg 35

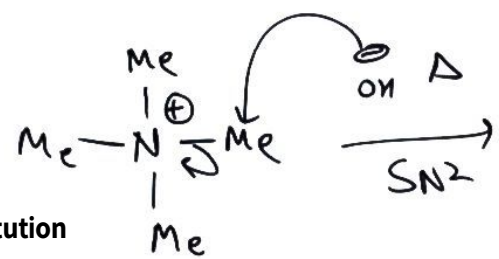



(i) MeI Δ excess
(ii) AgOH
(iii) Δ

(i) Me-I Δ
(ii) AgOH
(iii) Δ



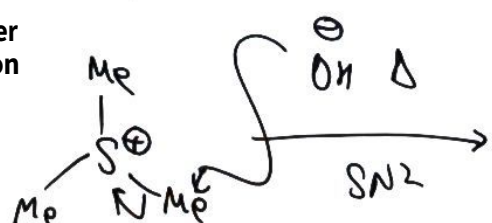

eg 36




$NMe_3$ + Me-OH
i.e.

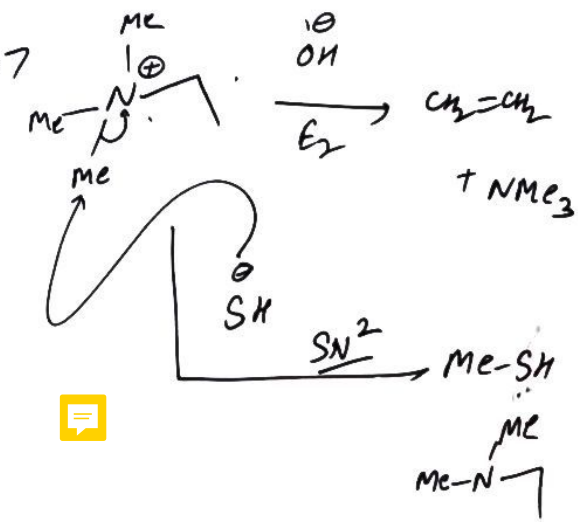
**if no β-H, then substitution**

**if β-H present, always prefer elimination over substitution**




$Me_2S$ + Me-OH

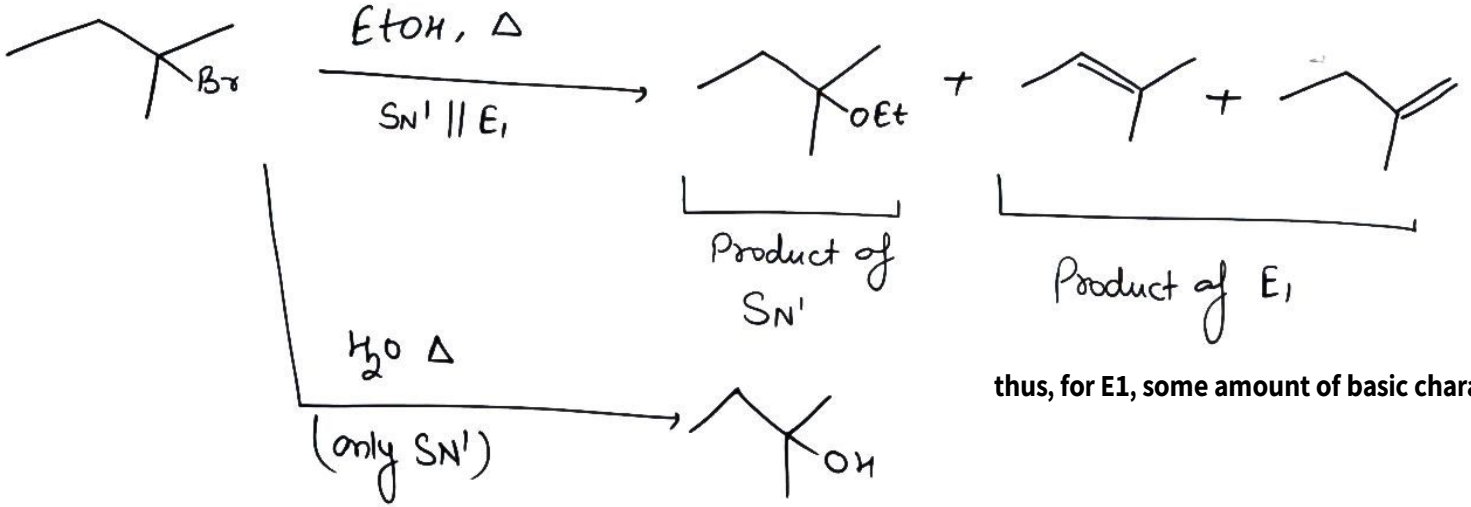
eg 37




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# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## (II) $E_1$ = Unimolecular elimination Reaction:-



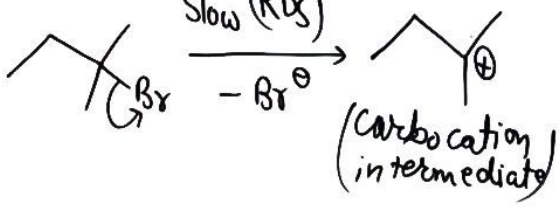

**thus, for E1, some amount of basic character is required**

(1) SN' || $E_1$ ( 1st step of mechanism (RDS) is same for both reaction)

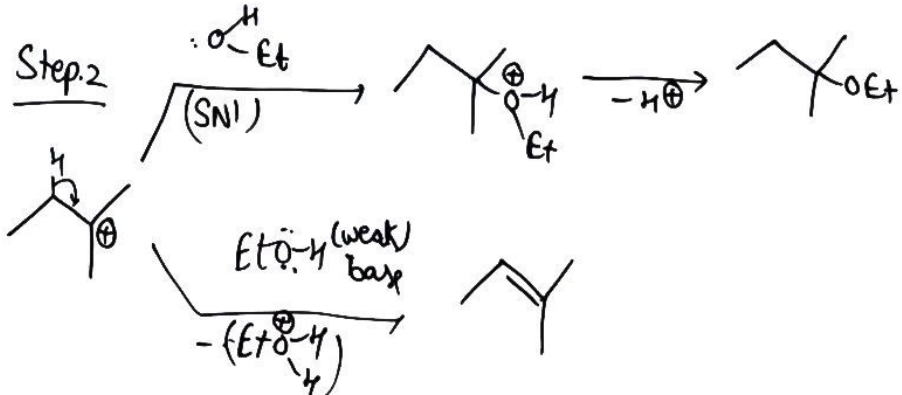
(2) $E_1$ ⇒ Two step mechanism.

(3) mechanism

Step.1




Step.2



# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## (II) $E_1$ = Unimolecular elimination Reaction:-

(3) Base = Weak bases are useful for $E_1$ elimination on high temp. (MeOH, Δ  EtOH, Δ etc)

(4) Rate of $E_1 \propto [R\text{-}X]^1 [Base]^0$

→ 1st order reaction

(5) Intermediate = Classical carbocation intermediate

(6) Reactivity of $E_1 \propto$ Stability of intermediate (form in RDS) not on rearranged carbocation

" " $3^\circ R\text{-}X > 2^\circ R\text{-}X > 1^\circ R\text{-}X$

(7) Rearrangement possible in intermediate carbocation.

→ major product forms by most stable carbocation.

(8) Major product = Saytzeff alkene.

(9) $E_1$ elimination is Non stereospecific reaction. not necessarily syn or anti, the H can be removed from any direction

(10) In $E_1$ elimination = KIE Not applicable.

(11) Solvent = Polar protic solvent

Note :- For alkyl halides generally

Product of $SN^1$ > product of $E_1$

on heating, % of E1 product will increase

(12) Reactivity of $E_1 \propto$ Stability of l.g

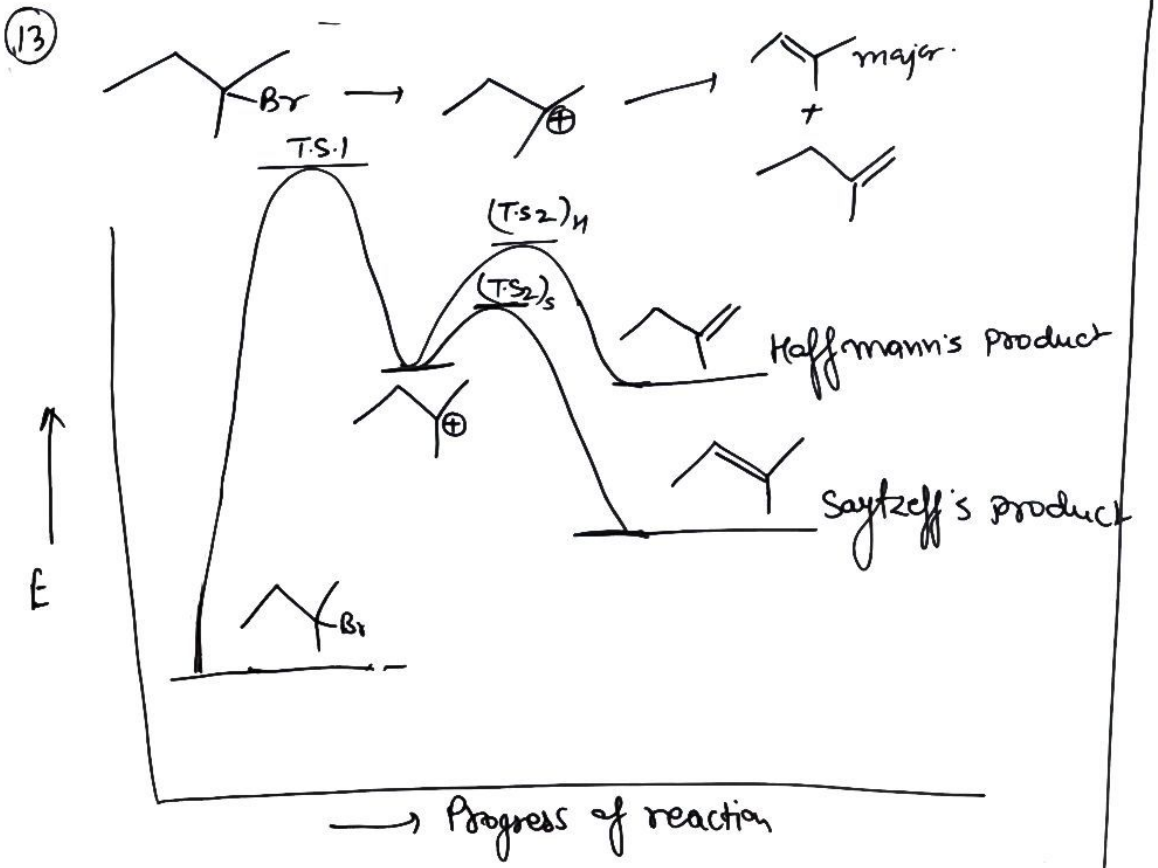
$R\text{-}I > R\text{-}Br > R\text{-}Cl > R\text{-}F$

↑ NO $E_1$ elimination

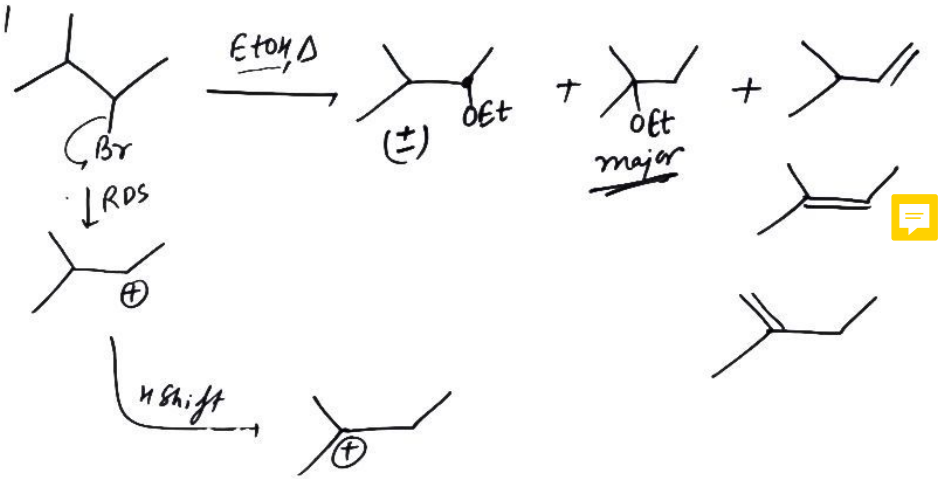
# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## (II) $E_1$ = Unimolecular elimination Reaction:-

(13)



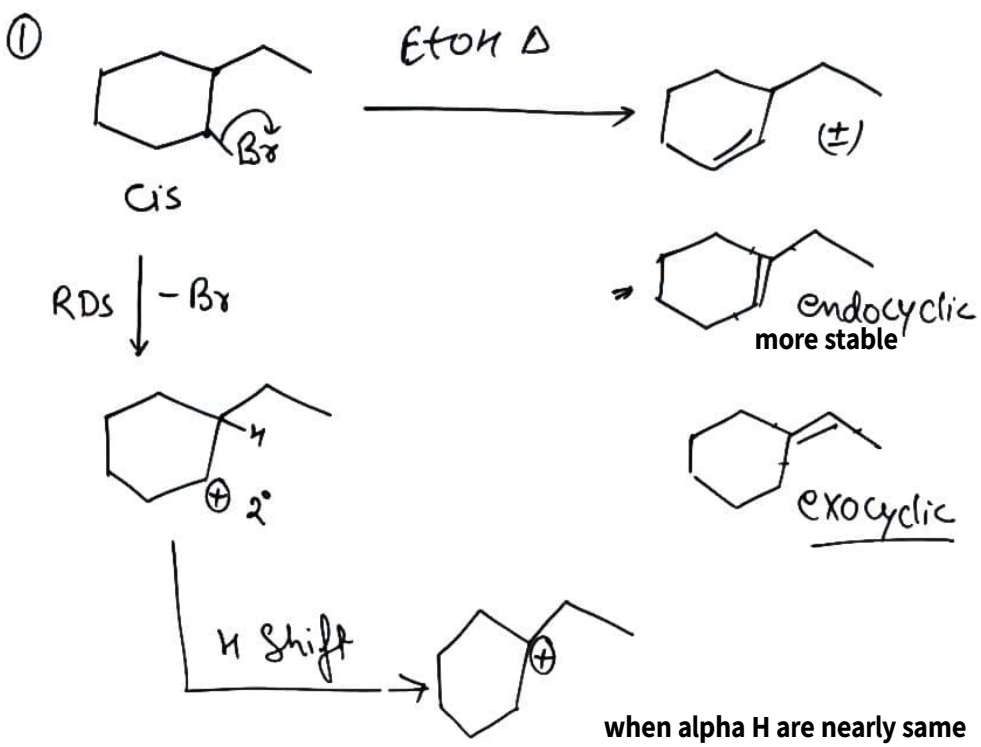

eg 1



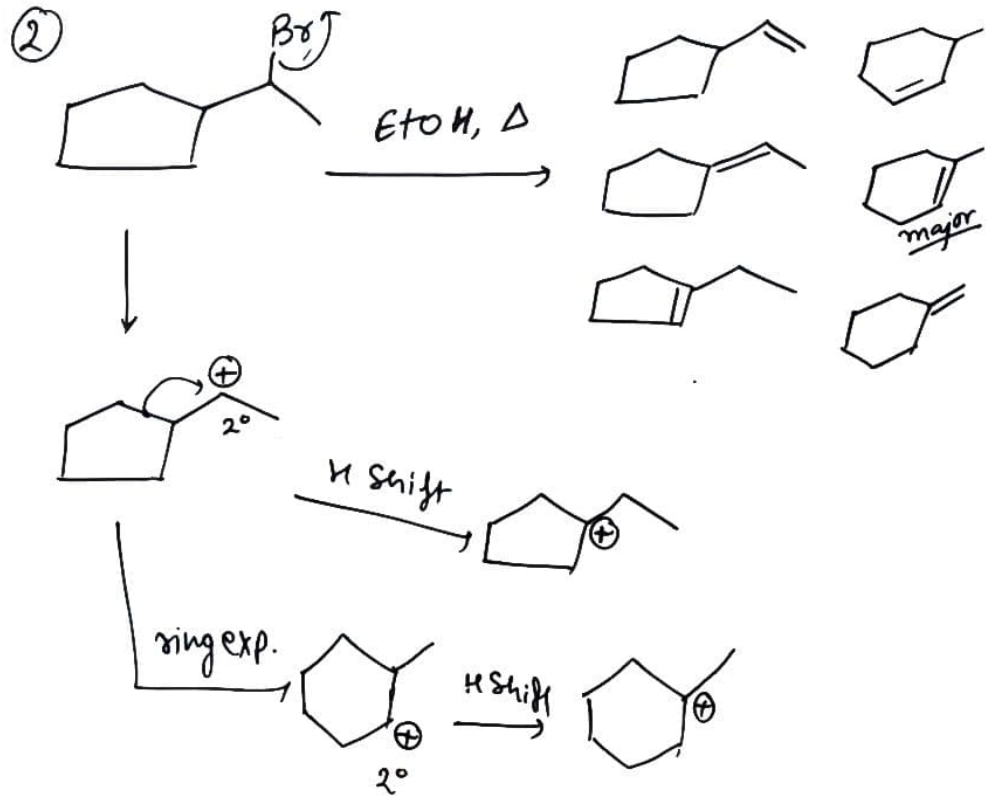
# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## (II) $E_1$ = Unimolecular elimination Reaction:-

**Ques.** Identify all possible elimination products.




when alpha H are nearly same endocyclic alkenes (in ring) are more stable than exocyclic alkene (outside ring)

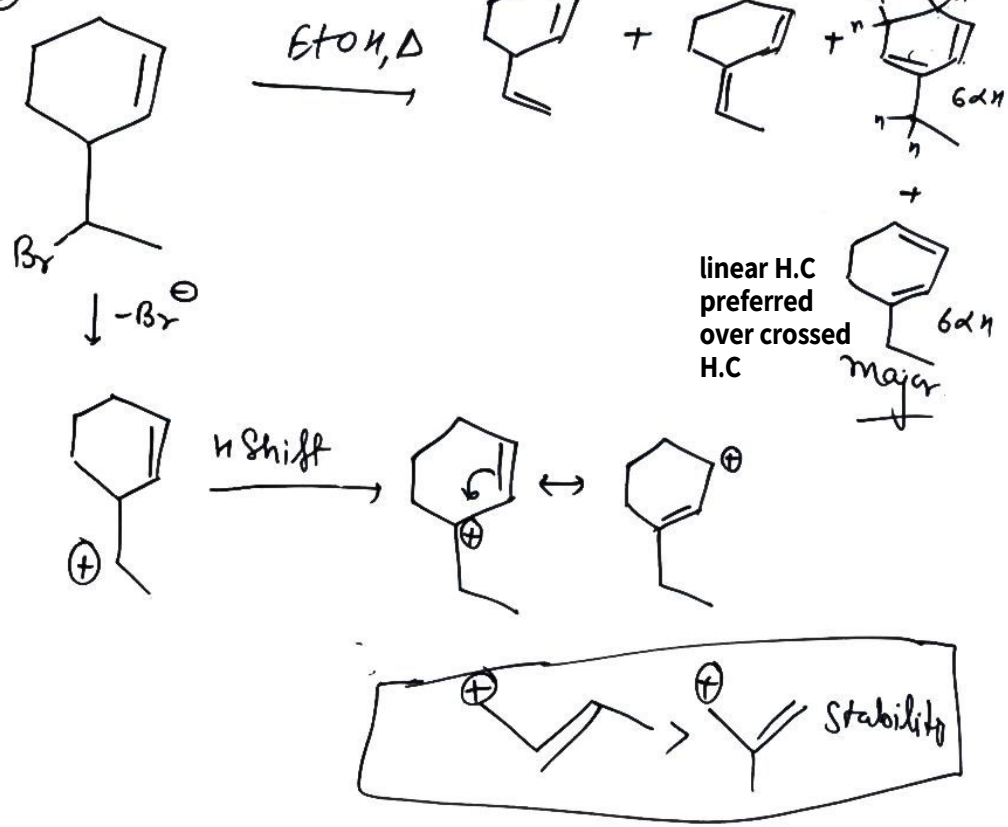


# ORGANIC REACTION MECHANISM - IV (ORM-IV)

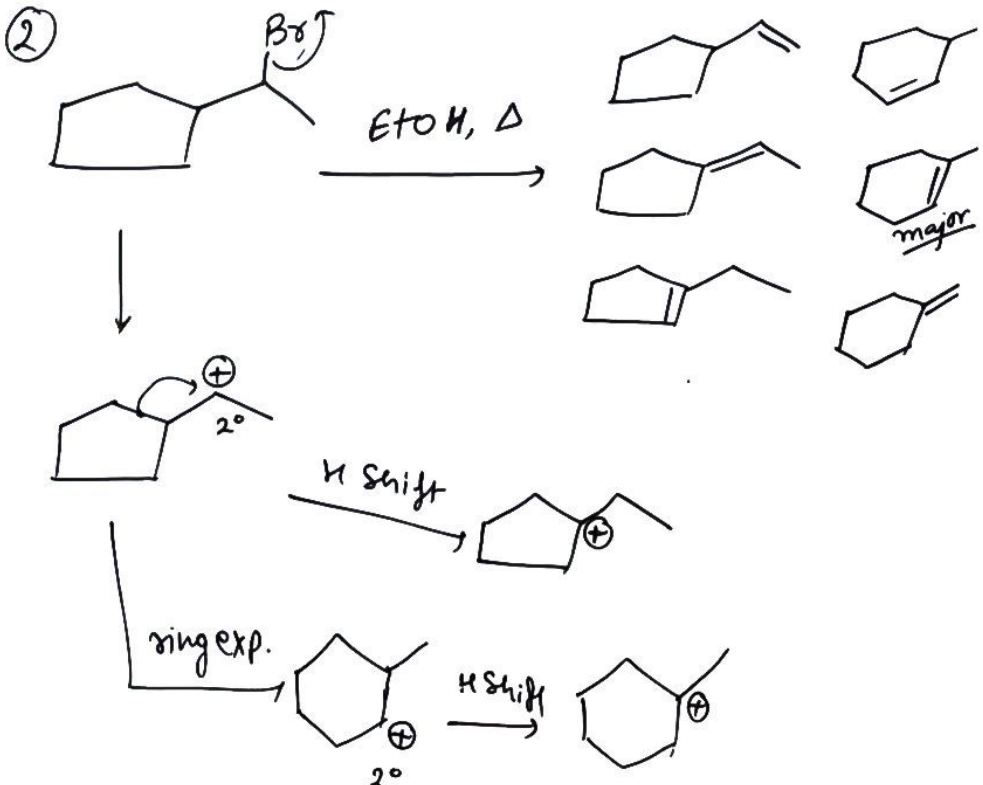
## (II) $E_1$ = Unimolecular elimination Reaction:-

Ques. Identify all possible elimination products.

③




②



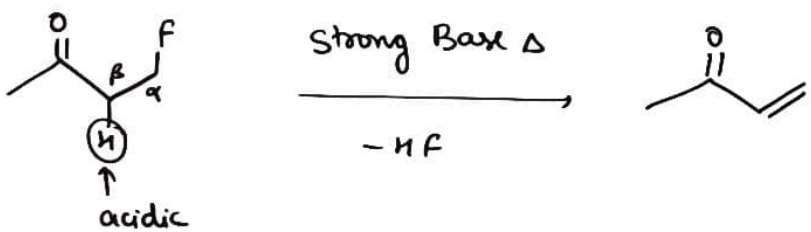
# ORGANIC REACTION MECHANISM - IV  ORM-IV

## (III) E1CB (Unimolecular Conjugate Base Elimination Reaction:-

favourable conditions ⇒ β. Hydrogen of leaving group should be acidic.

not necessary conditions

⇒ It is more favourable in presence of poor leaving groups. good lg prefer E2 over E1CB

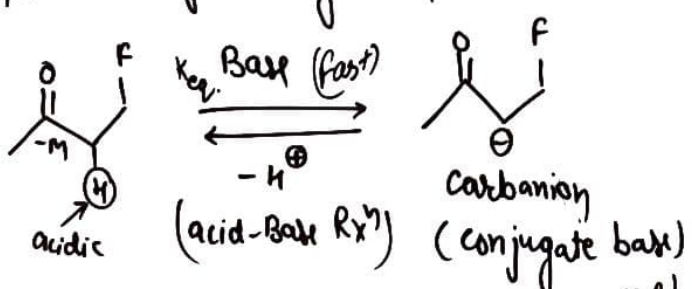



### mechanism

⇒ 2 step mechanism

⇒ in presence of strong base

Step-1




Step-2




⇒ RDS is unimolecular however the rex. is 2nd order

⇒ [Rate of Rxn = Rate of RDS] ∝ [C.B]

$$\Rightarrow K_{eq} = \frac{[C.B]}{[R\text{-}X][Base]} \quad ; \quad [C.B] = K_{eq}[R\text{-}X][Base]$$

⇒ Rate of Rxn $= K K_{eq} [R\text{-}X][Base]$

# ORGANIC REACTION MECHANISM - IV  ORM-IV

## (III) $E_{1cB}$ (Unimolecular Conjugate Base Elimination Reaction:-

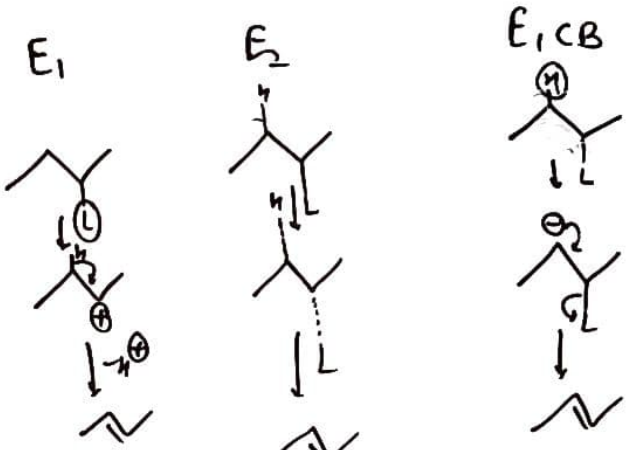
⇒ $E_{1cB}$ Reaction is 2nd order Rxn

⇒ Intermediate = carbanion

⇒ No Rearrangement in $E_{1cB}$

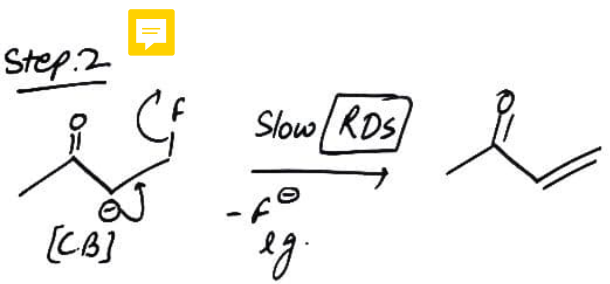
⇒ Non Stereospecific reaction

⇒ Kinetic Isotopic effect not applicable.




poor lg nikalne m nautanki krta hai, easily carbanion banke it pushes the lg out

Step.2




⇒ RDS is unimolecular

⇒ [Rate of Rxn = Rate of RDS] ∝ [C.B]

⇒ $K_{eq} = \frac{[C.B]}{[R\text{-}X][Base]}$ ; $[C.B] = K_{eq}[R\text{-}X][Base]$

⇒ Rate of Rxn = $K K_{eq}[R\text{-}X][Base]$

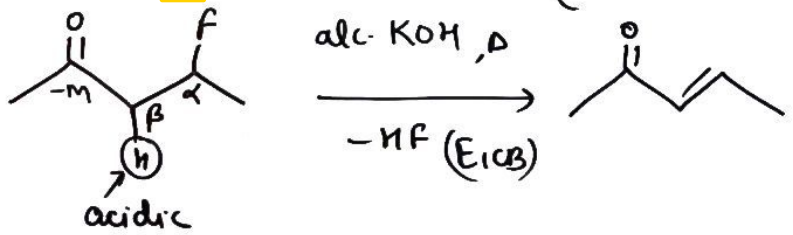
# ORGANIC REACTION MECHANISM - IV  ORM-IV

## (III) $E_{1cB}$ (Unimolecular Conjugate Base Elimination Reaction:-

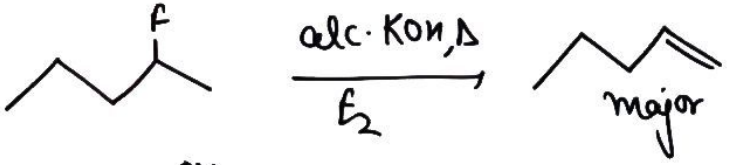
→ In $E_{1cB}$ reaction D. Exchange possible on β. Position of l.g. in presence of $\overset{\ominus}{O}D + D_2O$ or $RO^{\ominus} + ROD$. (due to formation of anion intermediate)

(due to its 1st step is ⇌ acid-base Rxn)

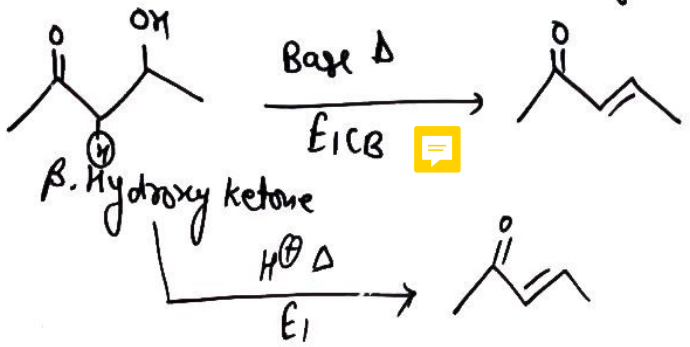
eg 1



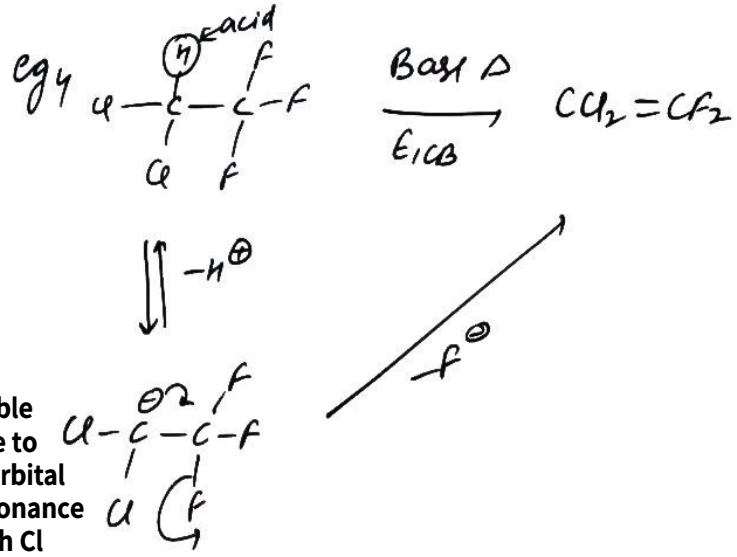

eg 2




eg 3




eg 4



# ORGANIC REACTION MECHANISM - IV  ORM-IV

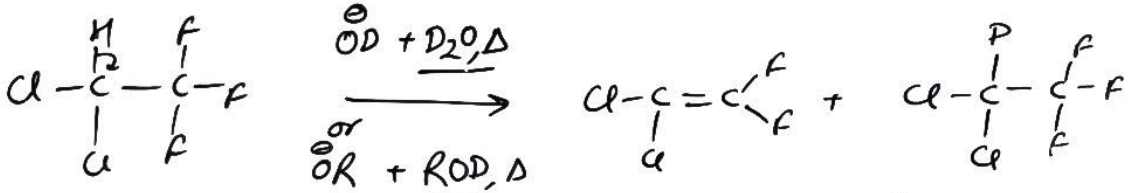
## (III) E1CB (Unimolecular Conjugate Base Elimination Reaction:-

$CH_2=CH-COOH$ acrylic acid

$CH_3-CH=CH-COOH$ Crotonic acid

$Ph-CH=CH-COOH$ Cinnamic acid

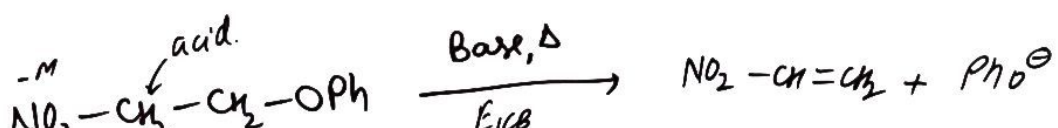
eg 5



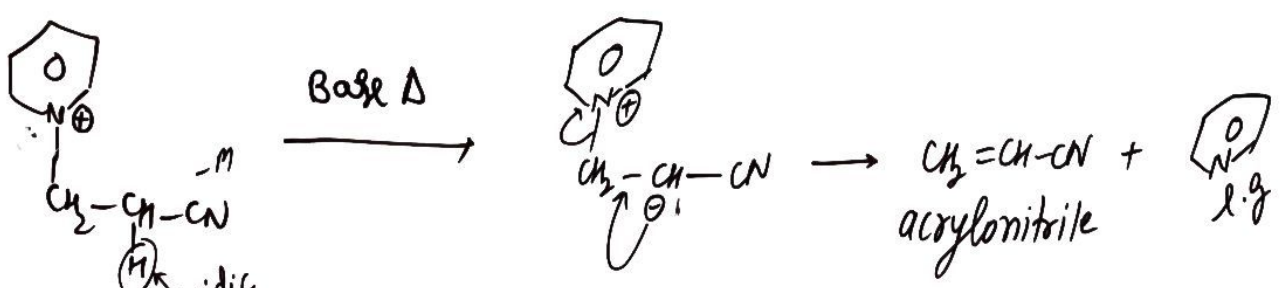
**formation of this product is proof for the formation of anion and hence E1CB mechanism**

**had this product not been formed, than the rex. would have been E2**

eg 6



eg 7

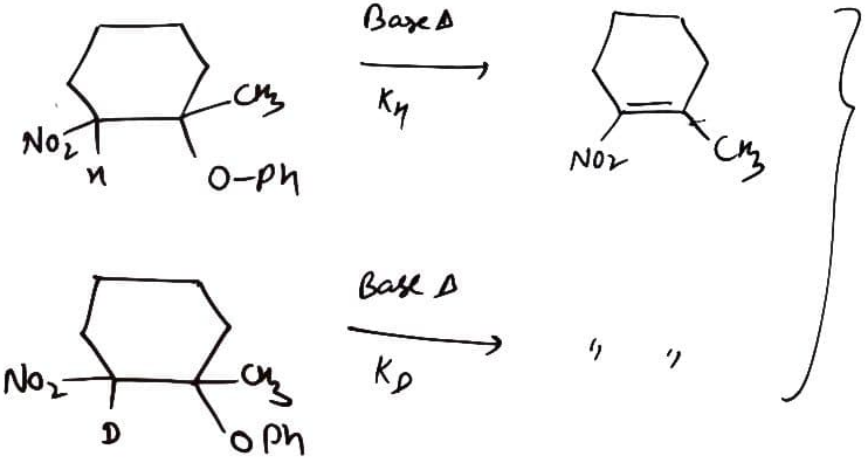


# ORGANIC REACTION MECHANISM - IV  ORM-IV

## (III) $E_{1CB}$ (Unimolecular Conjugate Base Elimination Reaction:-

**method of experimental identification**

eg 8

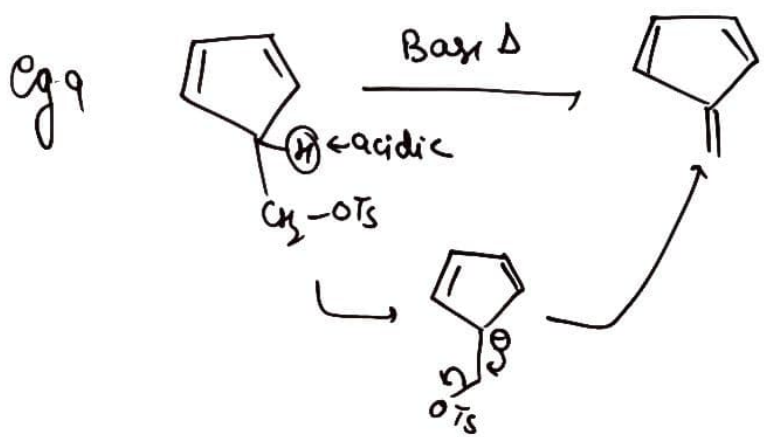



if $K_H \simeq K_D$ NO KIE ⇒ $E_{1CB}$

or

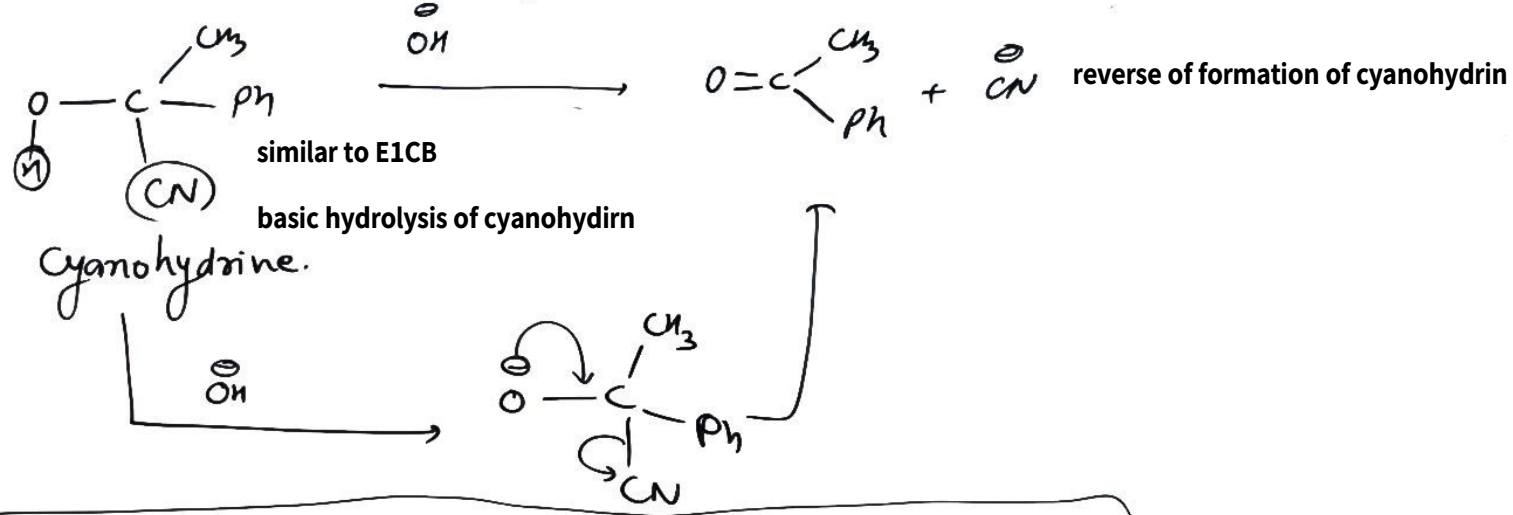
if $K_H > K_D$ KIE = ✓ ⇒ $E_2$

eg 9



## (III) E1CB (Unimolecular Conjugate Base Elimination Reaction:-

eg 10

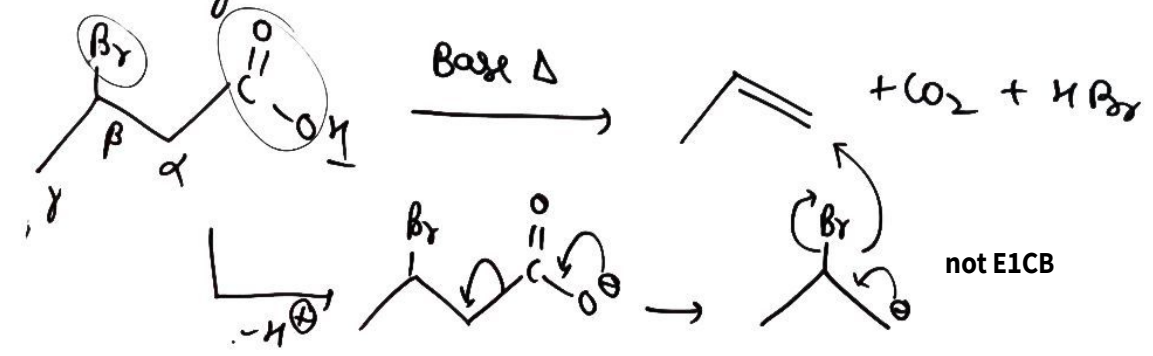



similar to E1CB

reverse of formation of cyanohydrin

basic hydrolysis of cyanohydirn

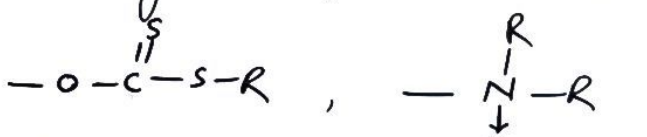
### Decarboxylation in β. Halocarboxylic acid.



# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## (IV) Ei (Internal elimination Reactions)

① It is possible in presence of specific leaving groups.

like,  ,  ,  etc

ester — Xanthate ester — Trialkyl amine oxide

② Only on high temp ( without base)

③ Cyclic transition state form = Pericyclic reaction

④ Stereospecific elimination.

⇒ Syn elimination. (CSE, TST)

⇒ T.S = Syn periplanar with eclipsed conformation.

# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## (IV) Ei (Internal elimination Reactions)

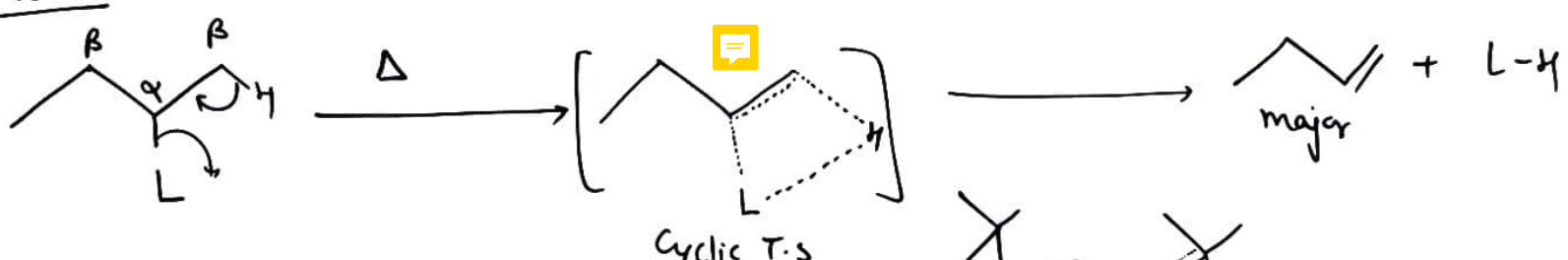
(5) No Rearrangement

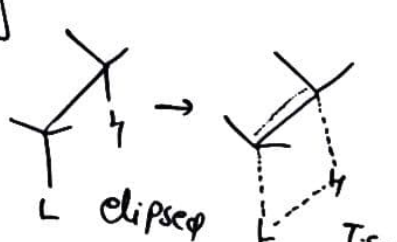
(6) Hoffmann's product = major product

( Lesser steric hindered cyclic T.S. are more stable)

**least sterically hindered H will always be removed in major product**

reaction







(7) One step mechanism

→ 1st order reaction.

→ Rate Ei α [R-L]

(8) kinetic Isotopic effect = applicable

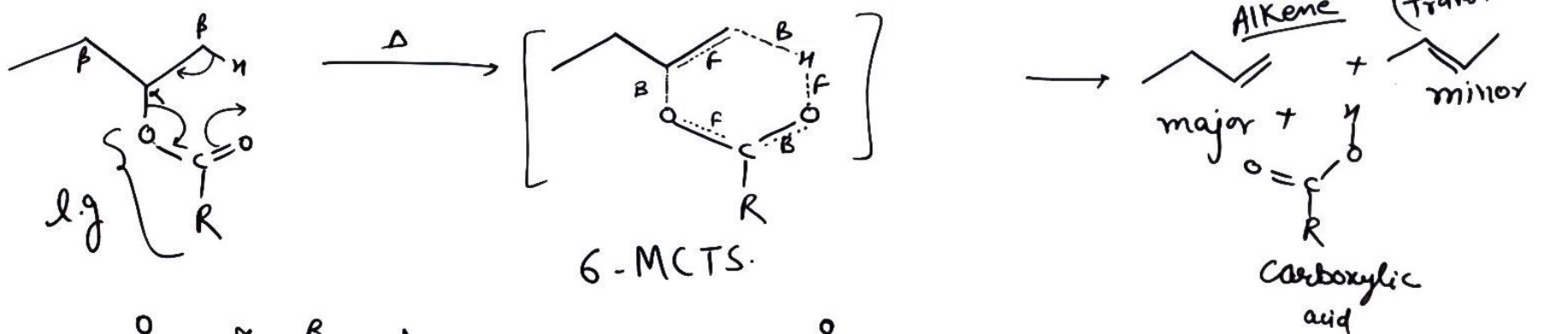
# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## (IV) Ei (Internal elimination Reactions)

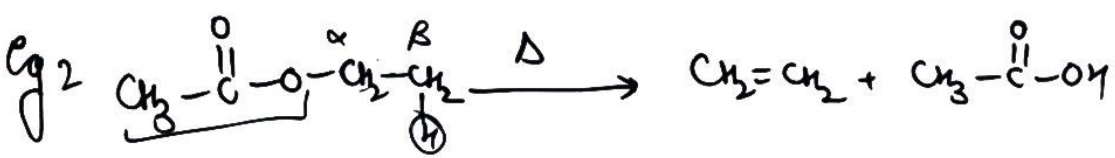
### (A) Pyrolysis of ester:-

**pyrolysis: dissociation of a compound into two or more compounds at high temperature i.e thermal dissociation**

eg1



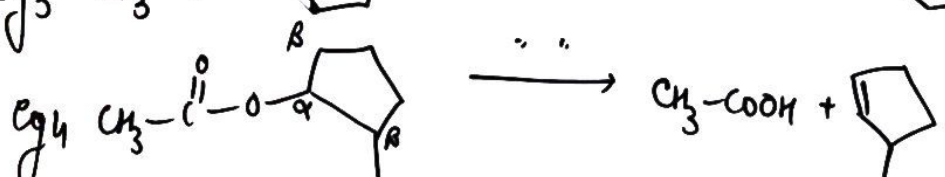

eg2




eg3




eg4



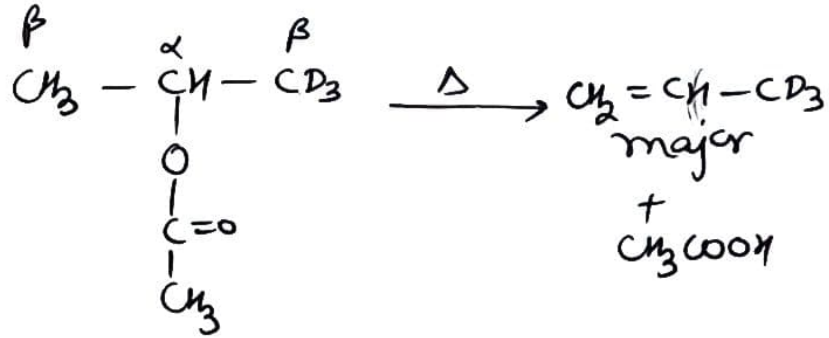
# ORGANIC REACTION MECHANISM - IV (ORM-IV)

CAT

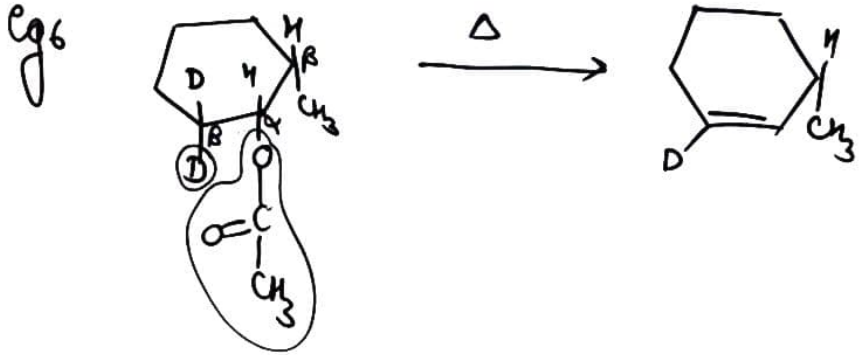
## (IV) Ei (Internal elimination Reactions)

### (A) Pyrolysis of ester:-

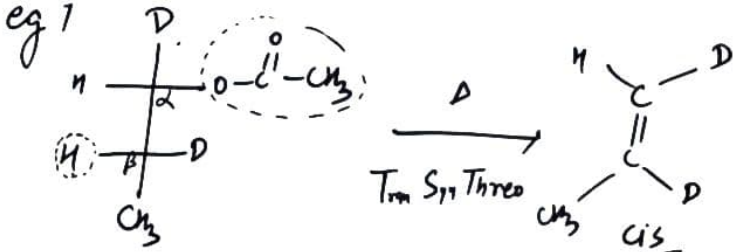
eg 5



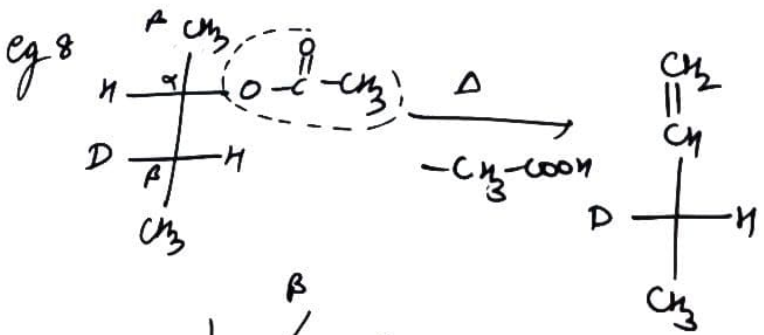

eg 6



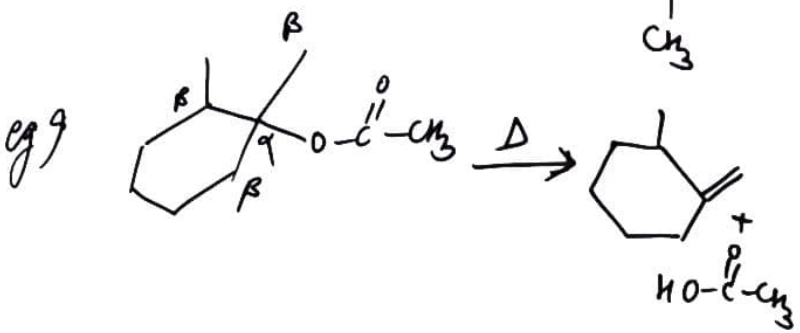

eg 7




eg 8

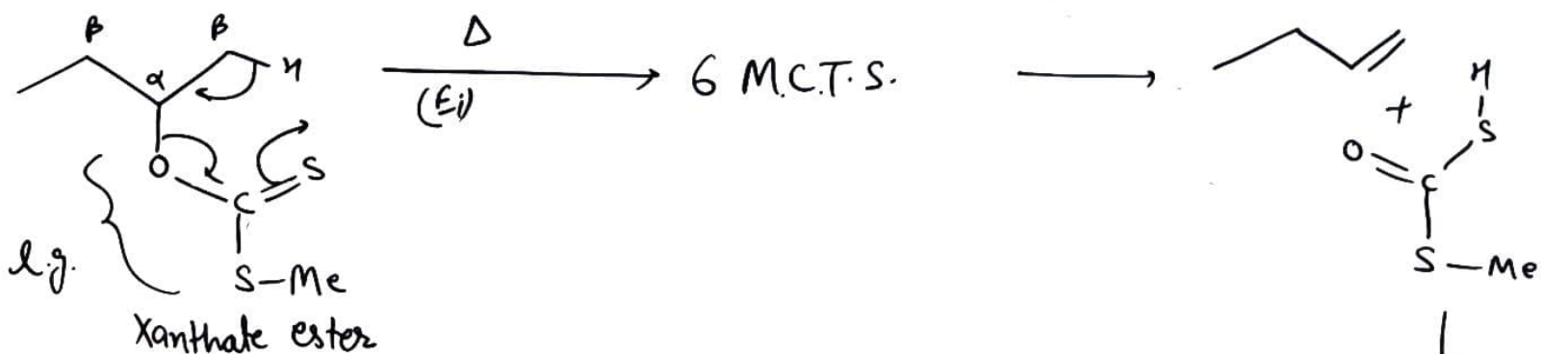



eg 9



## (IV) Ei (Internal elimination Reactions)

### (B) Pyrolysis of Xanthate ester (Chugaev Reaction)

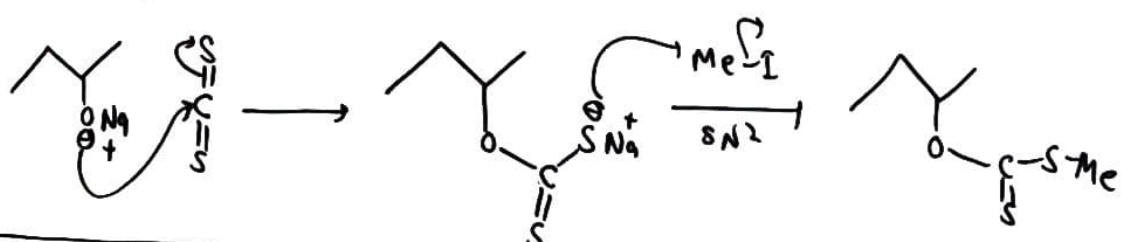



$O=C=S$

+

$Me-SH$

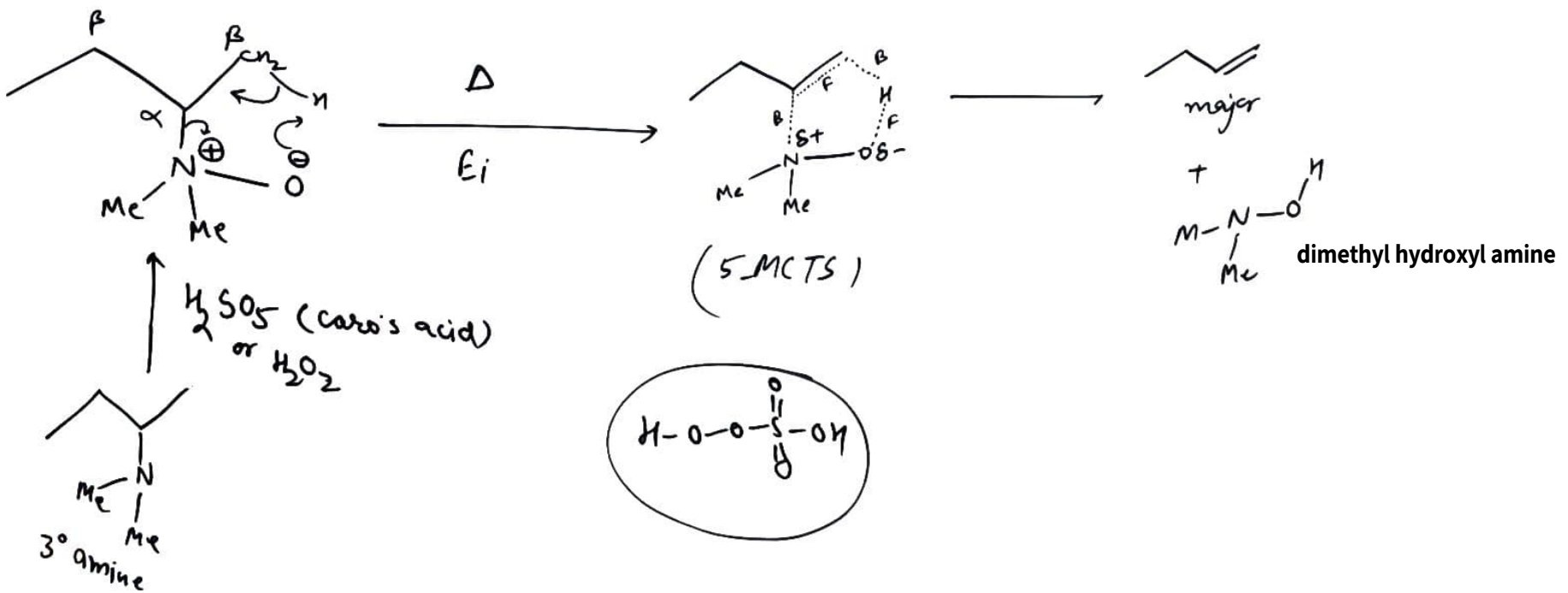
**Formation of Xanthate ester**



# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## (IV) Ei (Internal elimination Reactions)

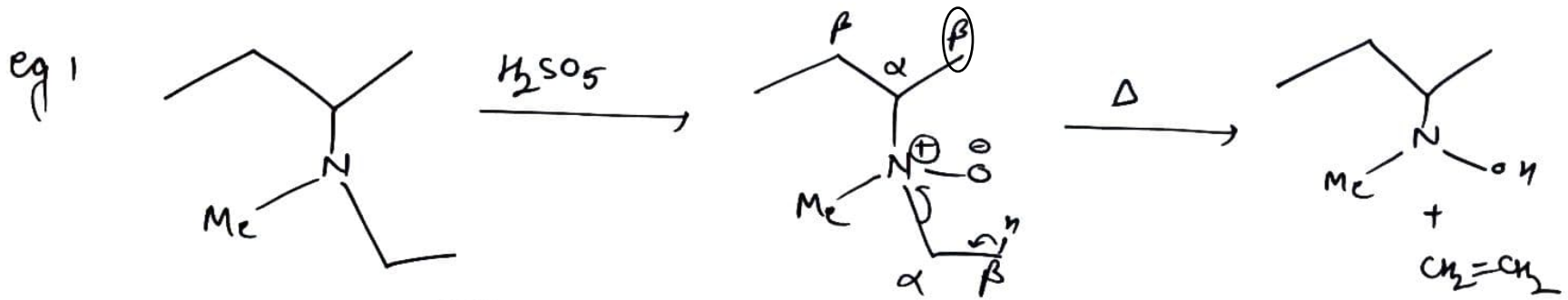
### (c) Pyrolysis in Tri alkyl amine oxide ( Cope elimination Rxn)

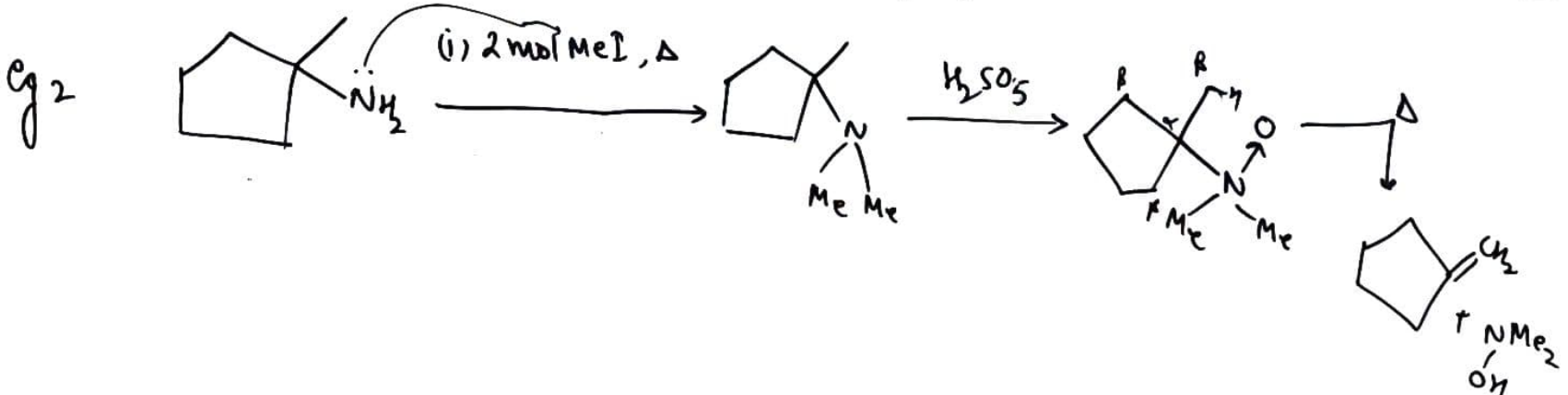


# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## (IV) Ei (Internal elimination Reactions)

### (c) Pyrolysis in Tri alkyl amine oxide ( Cope elimination Rxn)

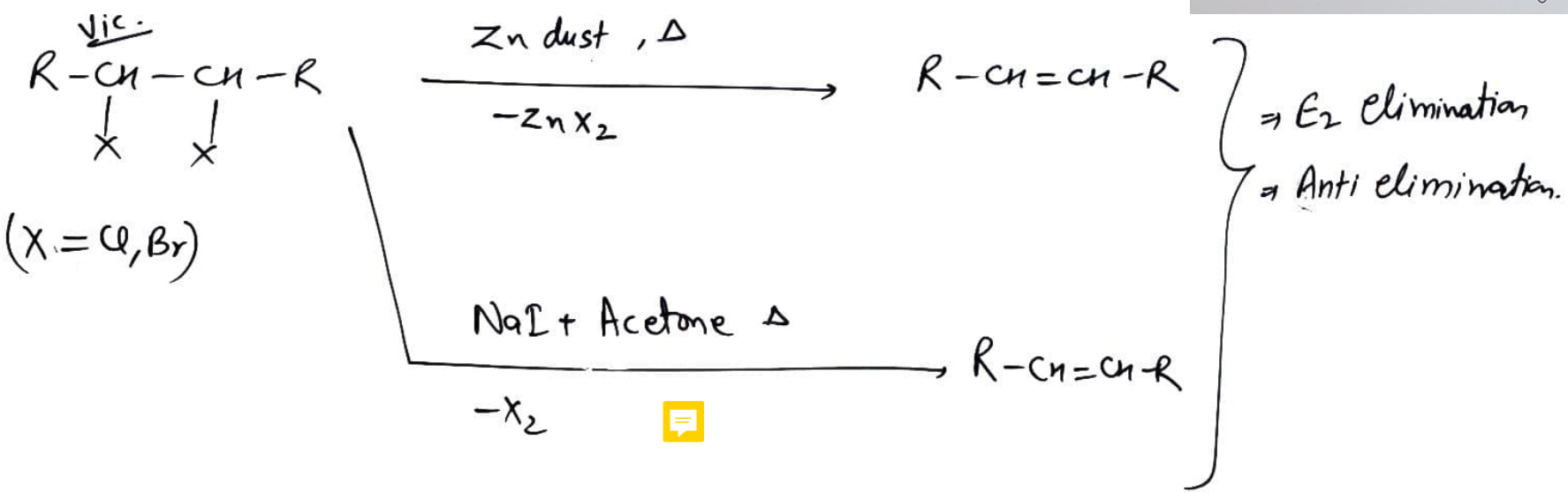


# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## Other elimination Reactions

### (a) Dehalogenation Reaction in vic. dihalides.




Other elimination Reactions

(a) Dehalogenation Reaction in vic. dihalides.

(b) Dehydration in alcohols

(c) 1.4-elimination = Conjugate elimination

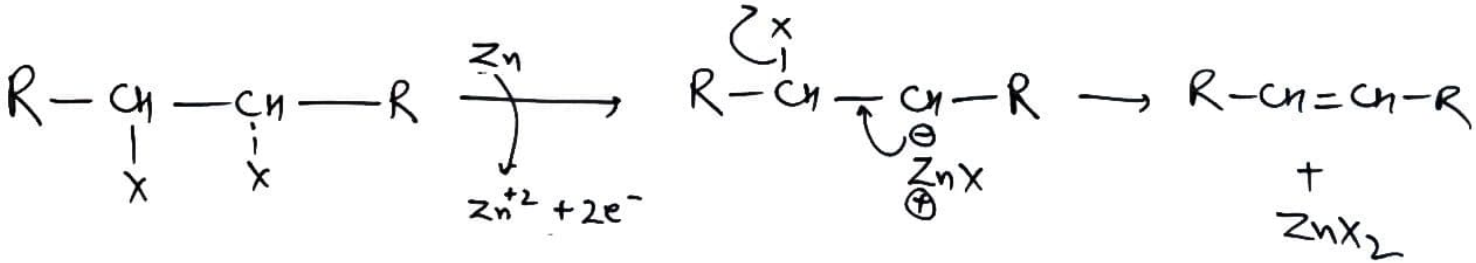
# ORGANIC REACTION MECHANISM - IV (ORM-IV)

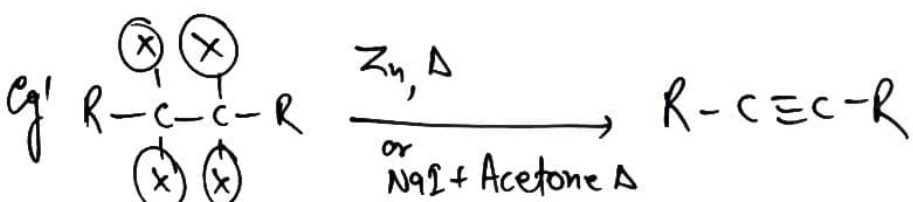
## Other elimination Reactions

### (a) Dehalogenation Reaction in vic. dihalides.

mech. for Zn. Δ

$$Zn \longrightarrow Zn^{+2} + 2e^-$$



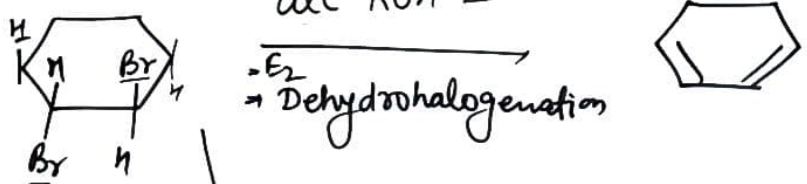
# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## Other elimination Reactions

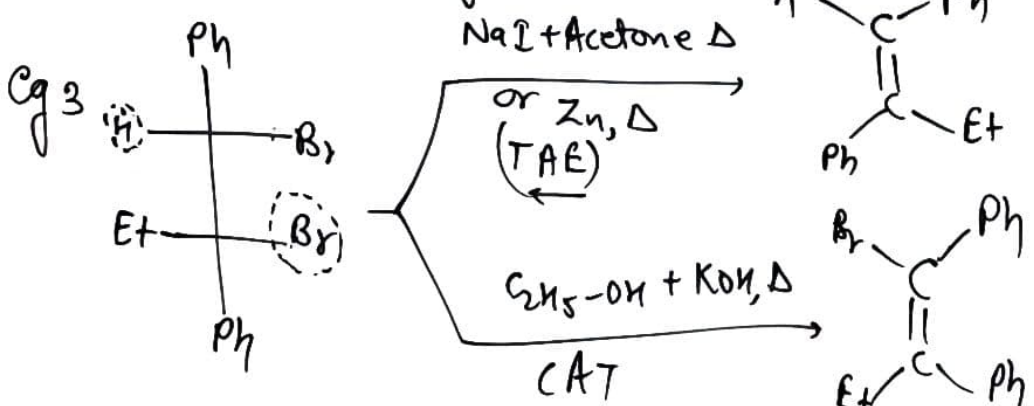
CAT
TAE

### (a) Dehalogenation Reaction in vic. dihalides.

Cg 2




Cg 3




(Dehalogenation)

(Dehydrohalogenation)

# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## Other elimination Reactions

X can be iodine here

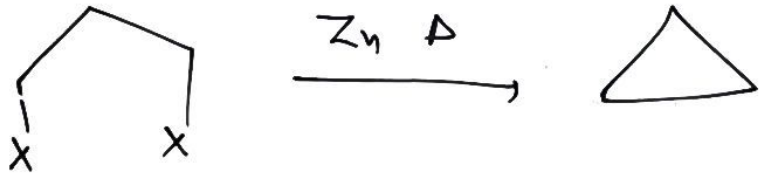
### Other Reaction of Zn

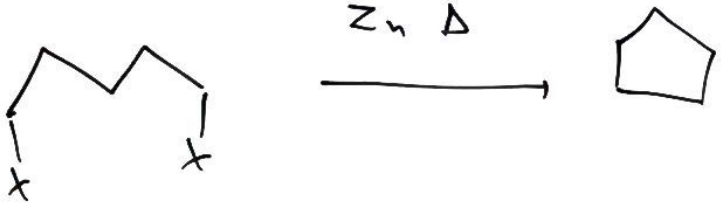
① $R-X \xrightarrow{Zn,\ \Delta} R-R$

X ≠ F

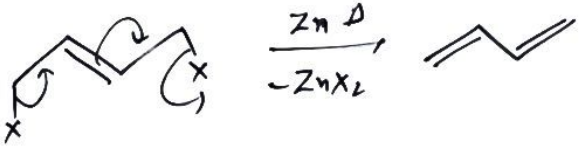
$Zn \rightarrow Zn^{+2} + 2e^-$

$[R^{\delta-}-Zn-X]$ Frankland's reagent

② 

③ 

even six and seven member ring forming reactions have also been observed

④ 

⑤ $R-CH(X)(X)$ gem. $\xrightarrow{Zn\ \Delta} R-C\equiv C-R$

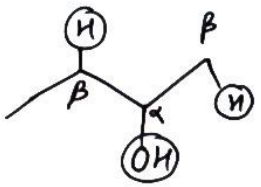
⑥ $R-C(X)(X)(X) + 6Ag + X-C(X)(X)-R \rightarrow -6AgX\downarrow$

$R-C\equiv C-R$

» $CHI_3 \xrightarrow[\Delta]{Ag} H-C\equiv C-H$

# ORGANIC REACTION MECHANISM - IV (ORM - IV)

## (b) Dehydration of alcohols:-




dehydrating agent, Δ

$-H_2O$



S.A. (Trans + cis)

+



H.A.

### Dehydrating agent

① Conc. $H_2SO_4$ Δ
② Conc. $H_3PO_4$, Δ

⇒ mainly $E_1$ elimination
⇒ Intermediate = Carbocation
⇒ Rearrangement possible
⇒ Saytzeff's alkene = major product

③ $P_2O_5$, Δ
④ $Al_2O_3$, Δ
⑤ $POCl_3 + Py$, Δ

⇒ $E_2$ elimination
⇒ T.S.
⇒ No Rearrangement
⇒ Saytzeff's product = major

⑥ $ThO_2$, Δ
⑦ $NO_2$, Δ

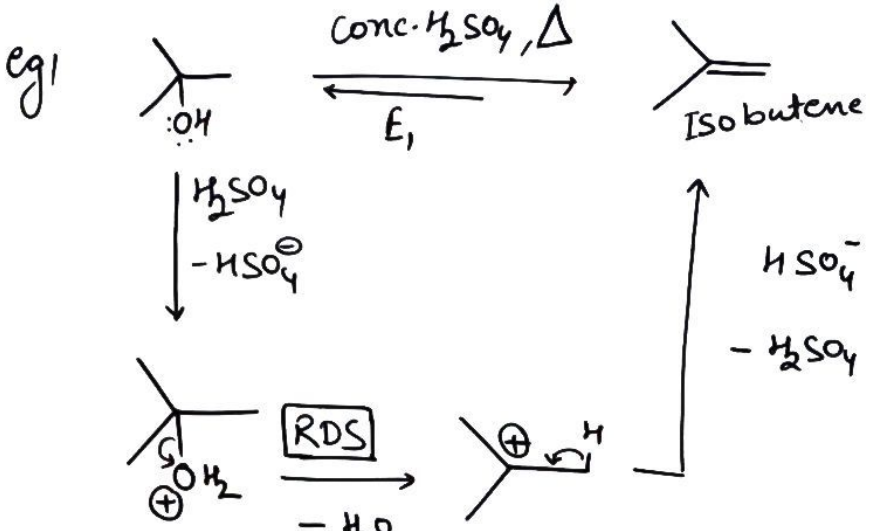
⇒ $E_2$ elimination
⇒ cyclic T.S.
⇒ No Rearrang.
⇒ Hoffmann product = major.

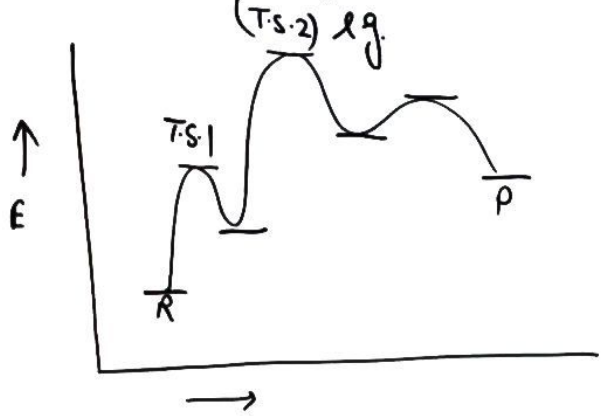
due to the large size

# ORGANIC REACTION MECHANISM - IV (ORM - IV)

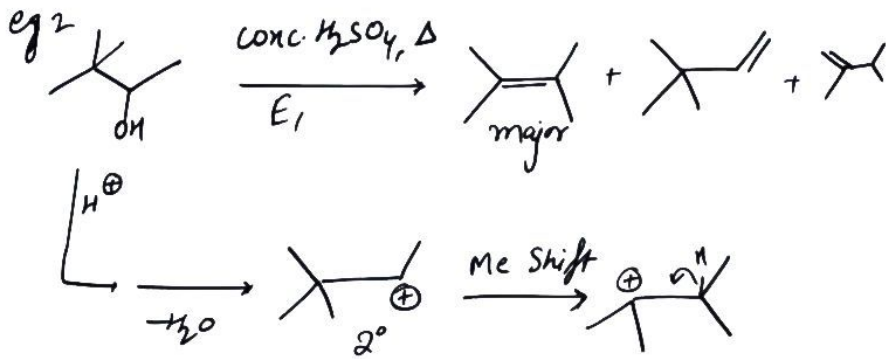
## (b) Dehydration of alcohols:-

eg 1




eg 2




Note ⇐ ⇒ (3°, 2°, β-crowded benzylic, allylic alcohol) $\xrightarrow[\Delta]{\text{conc. } H_2SO_4} E_1$

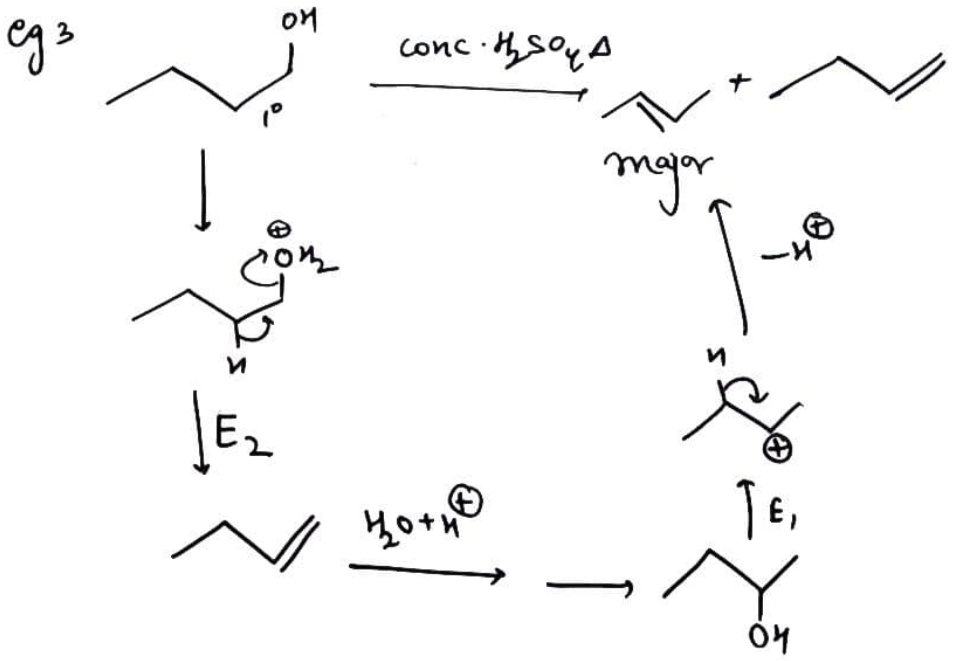
⇒ 1° alcohol $\xrightarrow{\text{Conc. } H_2SO_4} E_2$

**lekin major product**
**E1 wala hi banana**

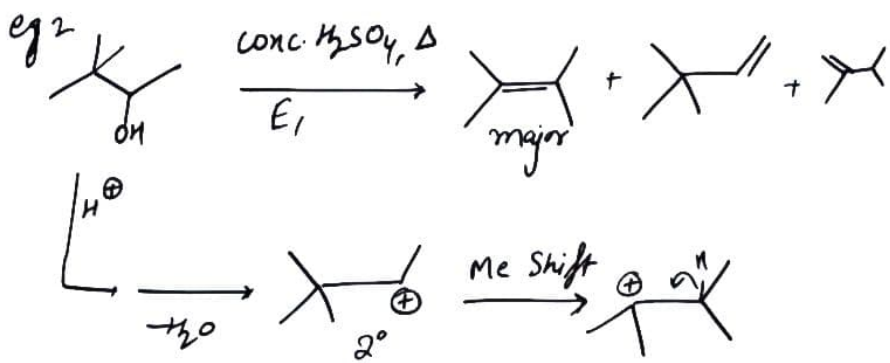
# ORGANIC REACTION MECHANISM - IV (ORM - IV)

## (b) Dehydration of alcohols:-

eg 3




eg 2




Note ⇐

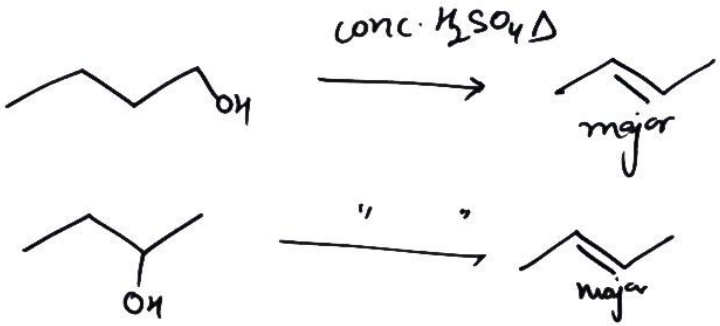
⇒ (3°, 2°, β-crowded benzylic, allylic alcohol) $\xrightarrow[\Delta]{\text{conc. } H_2SO_4}$ $E_1$

⇒ 1° alcohol $\xrightarrow{\text{Conc. } H_2SO_4}$ $E_2$

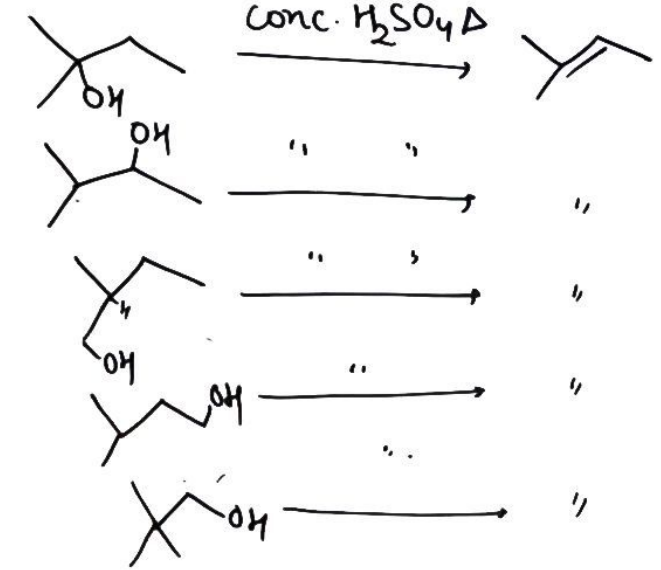
# ORGANIC REACTION MECHANISM - IV (ORM - IV)

## (b) Dehydration of alcohols:-

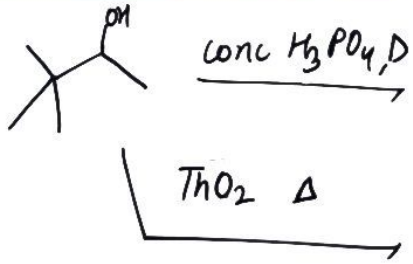
eg 4



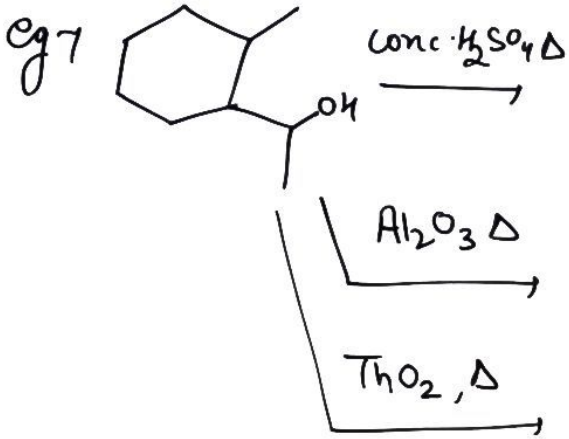

eg 5




eg 6



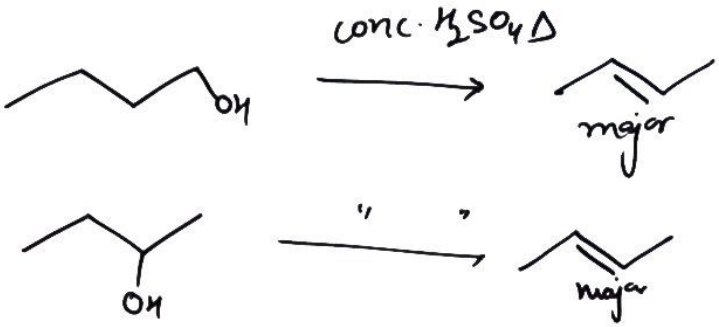

eg 7



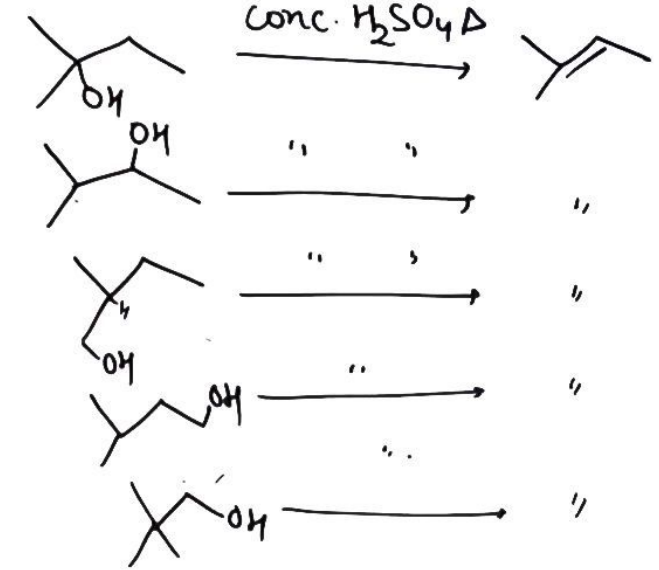
# ORGANIC REACTION MECHANISM - IV (ORM - IV)

## (b) Dehydration of alcohols:-

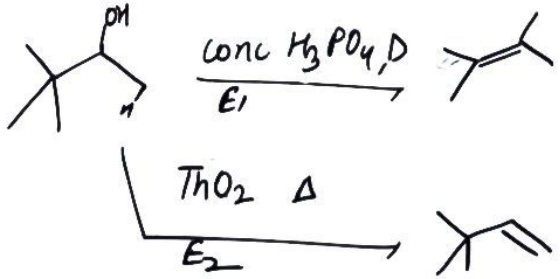
eg 4



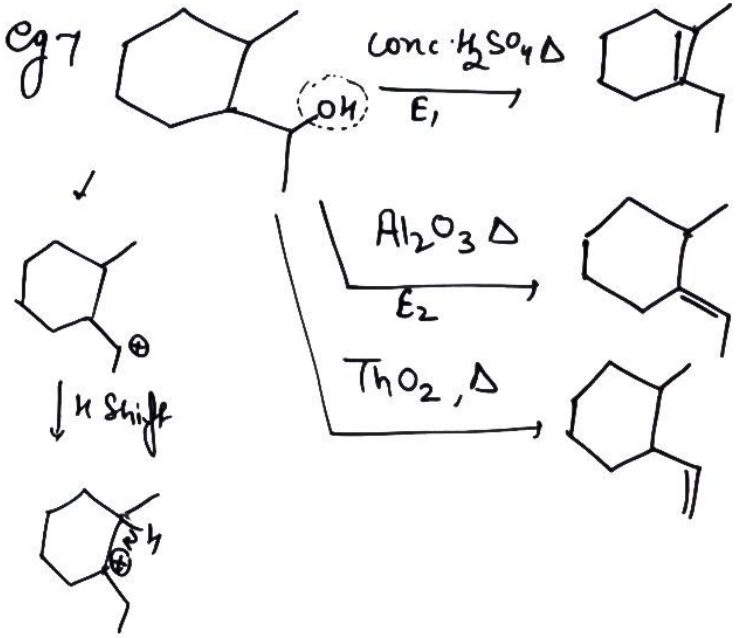

eg 5




eg 6



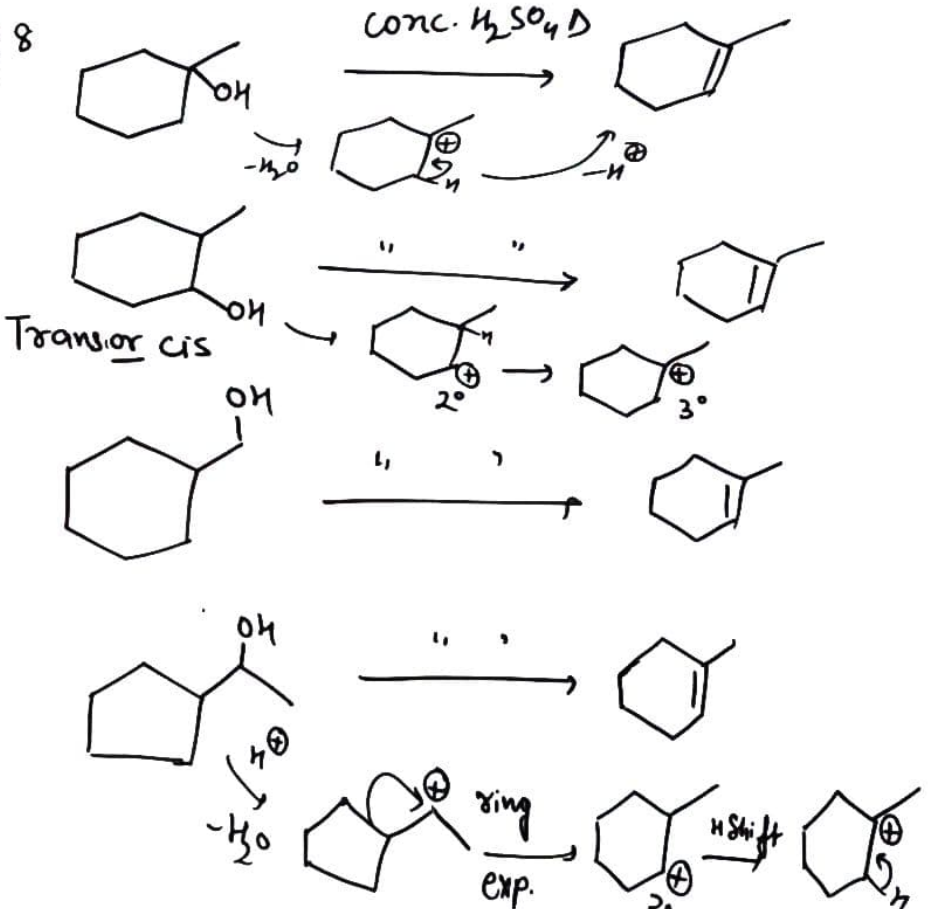

eg 7



# ORGANIC REACTION MECHANISM - IV (ORM - IV)

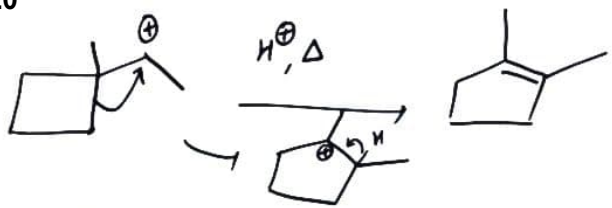
## (b) Dehydration of alcohols:-

eg 8

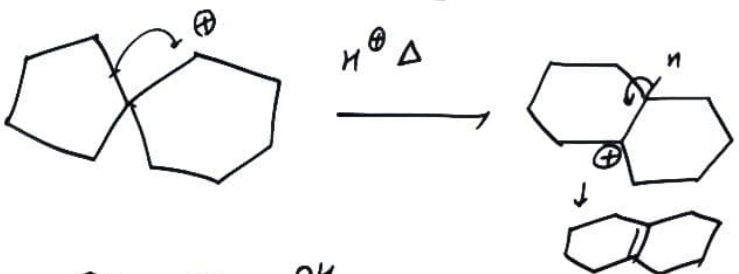



in eg9 and eg10 -OH in place of +

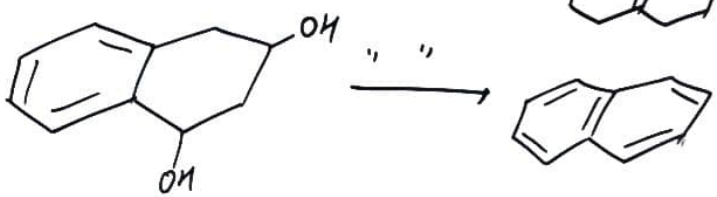
eg 9




eg 10



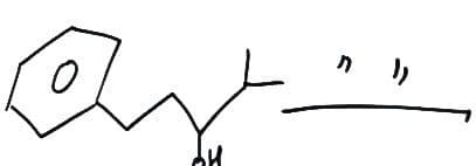

eg 11




eg 12



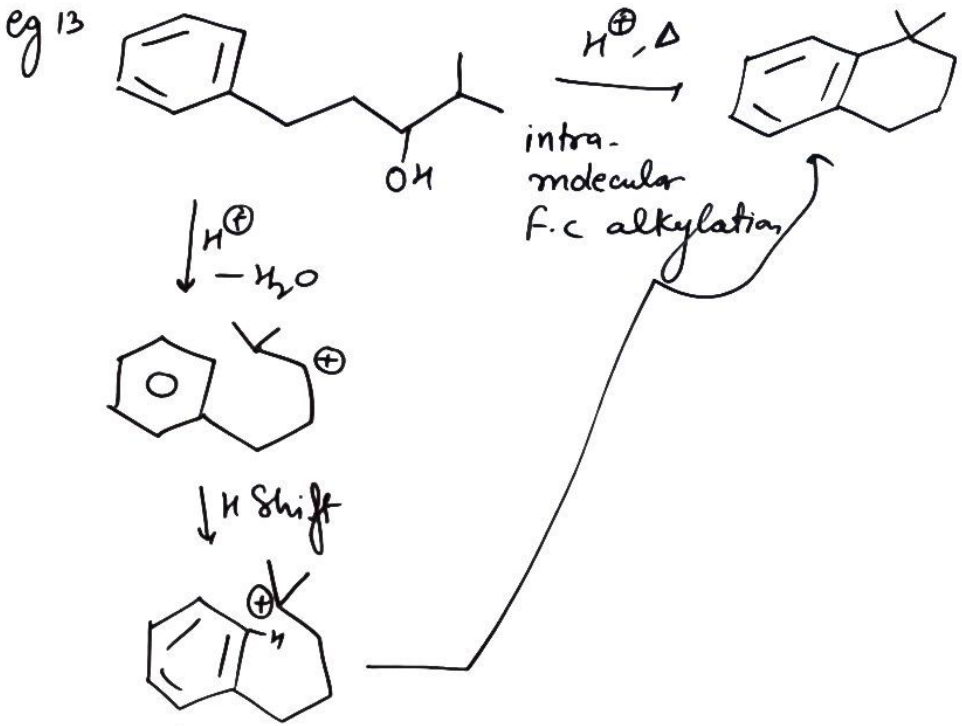

eg 13



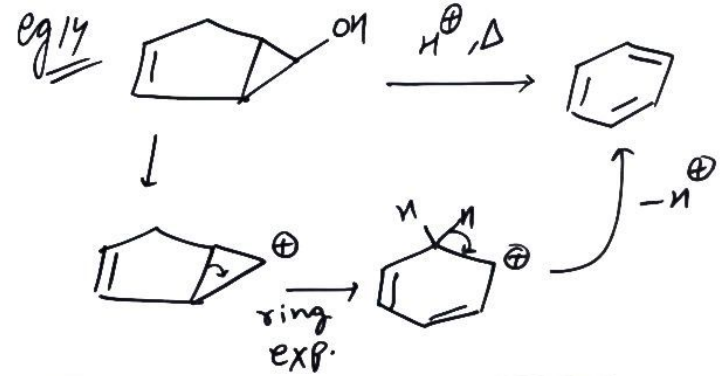
# ORGANIC REACTION MECHANISM - IV (ORM - IV)

## (b) Dehydration of alcohols:-

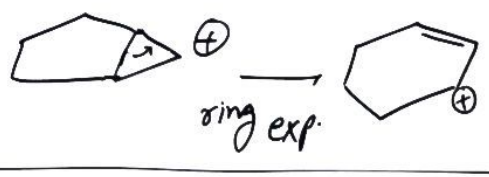
eg 13



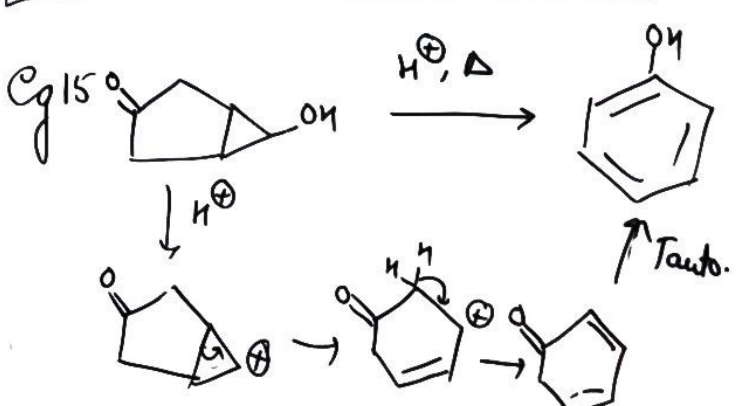

eg 14




Hint



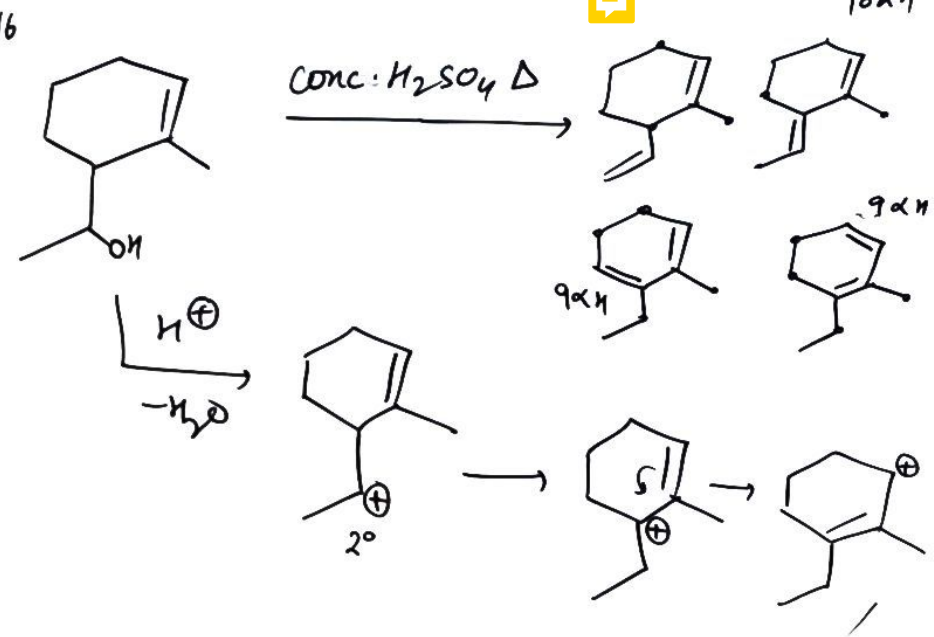

Eg 15



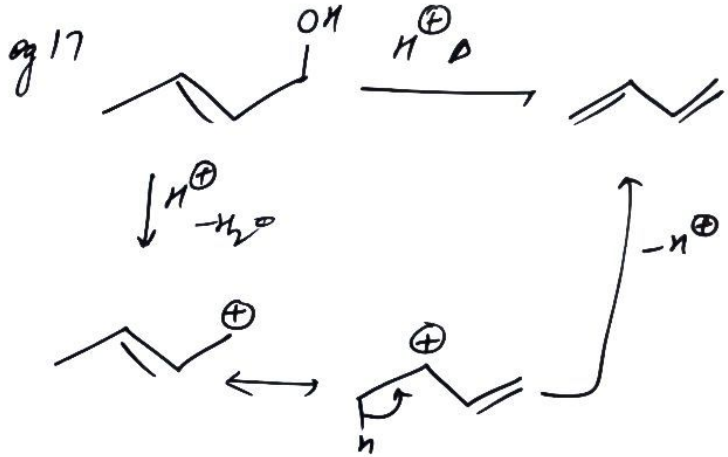
# ORGANIC REACTION MECHANISM - IV (ORM - IV)

## (b) Dehydration of alcohols:-

eg 16




eg 17




eg 18



# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## (d) Dehydration of alcohols :-

eg 19 ★

$CH_3-CH_2-OH \xrightarrow[0^\circ C]{\text{conc. } H_2SO_4} CH_3-CH_2-\overset{\oplus}{O}H_2\,]\,HSO_4^{\ominus}$

$CH_3-CH_2-OH \xrightarrow[\sim 100^\circ C]{\text{Conc. } H_2SO_4} CH_3-CH_2-O-\overset{O}{\overset{\|}{S}}(=O)-OH$

ethyl hydrogen sulphate

laboratory method of ether synthesis: Williamson's Continuous Ether Synthesis

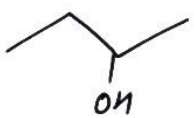
$CH_3-CH_2-OH \xrightarrow[140^\circ C]{\text{Conc. } H_2SO_4} C_2H_5-O-C_2H_5$

$CH_3-CH_2-OH \xrightarrow{H^{\oplus}} CH_3-\overset{\oplus}{C}H_2 \xrightarrow[SN1]{C_2H_5-\ddot{O}H} \xrightarrow{-H^{\oplus}} C_2H_5-O-C_2H_5$ (with $-H_2O$)

$CH_3-CH_2-OH \xrightarrow[160^\circ C]{\text{Conc. } H_2SO_4} CH_2=CH_2$ ($E_1$)

eg 20

$CH_3-CH_2-OH \xrightarrow[160^\circ C]{95\%\ H_2SO_4} CH_2=CH_2$




$\xrightarrow[\sim 100^\circ C]{60\%\ H_2SO_4}$

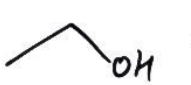





$\xrightarrow[80^\circ C]{20\%\ H_2SO_4}$



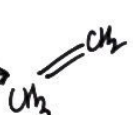
eg 21




$\xrightarrow[250^\circ C]{Al_2O_3}$

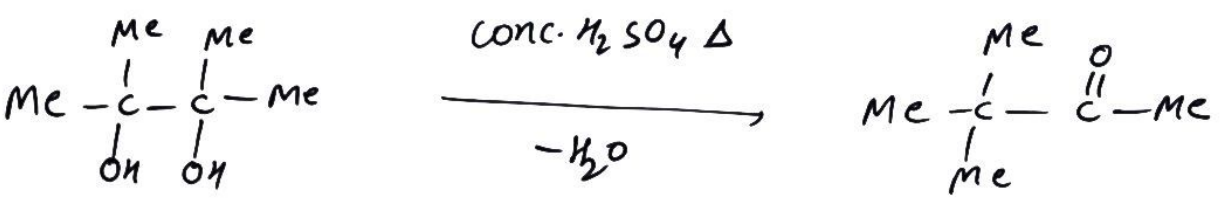


$\xrightarrow[380^\circ C]{Al_2O_3}$



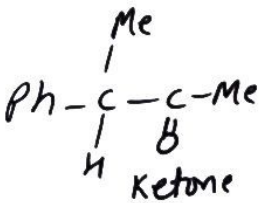
# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## Pinacol - Pinacolone Rearrangement:-

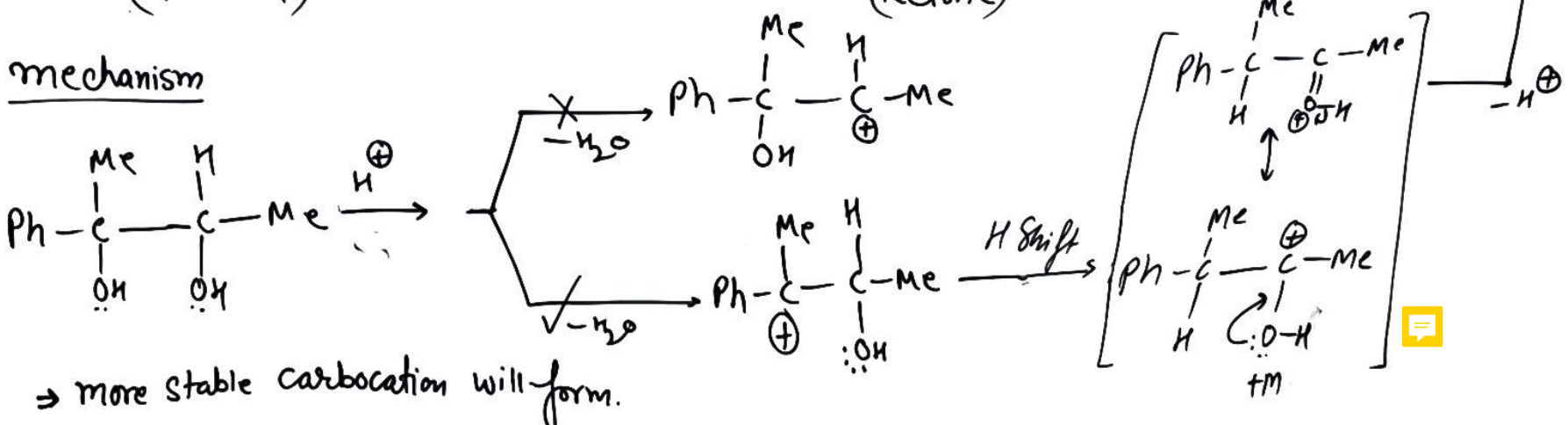



Pinacol (vic. diol)

Pinacolone (Ketone)




**mechanism**




⇒ more stable carbocation will form.

$-\ddot{O}H > Ph$ +M

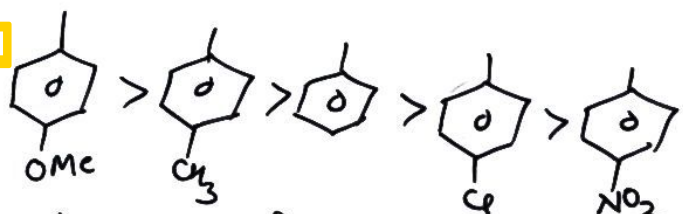
# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## Pinacol - Pinacolone Rearrangement:-

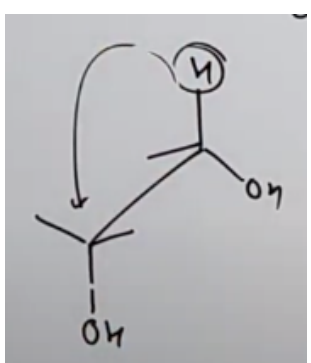
Generally ⇒ H > D > Ph > alkyl

⇒ migratory apptitute ⇒ Ph > H > D > Alkyl

⇒ " " in alkyl groups ⇒ $3^\circ > 2^\circ > 1^\circ > CH_3$

⇒ " " in aryl group $\alpha$ $\pi e^-$ density ⇒ 

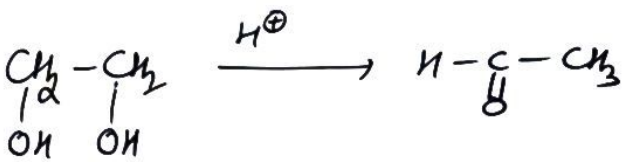

⇒ migration of group from opp. side (anti) w.r.t. $H_2O$ leaving group will shift.



# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## Pinacol - Pinacolone Rearrangement:-

eg 1



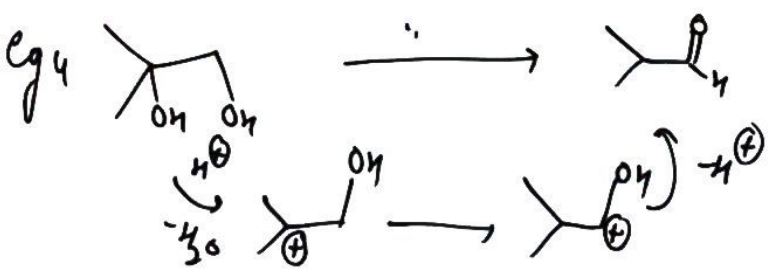
eg 2



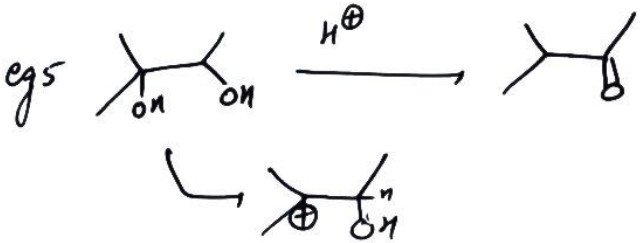
eg 3



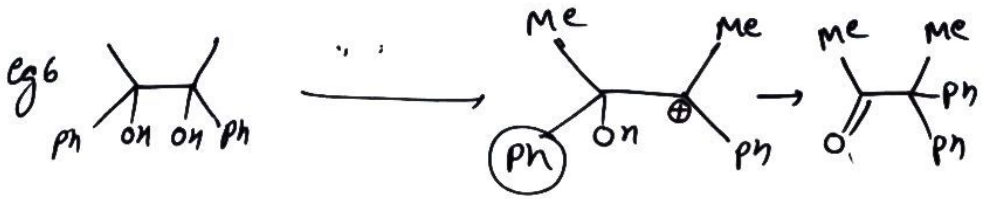
eg 4



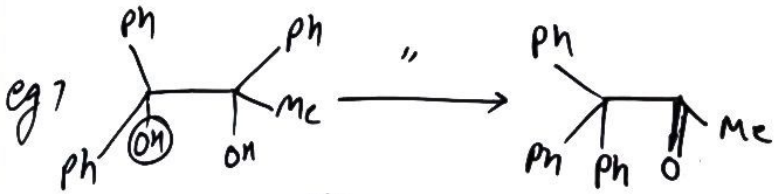
eg 5



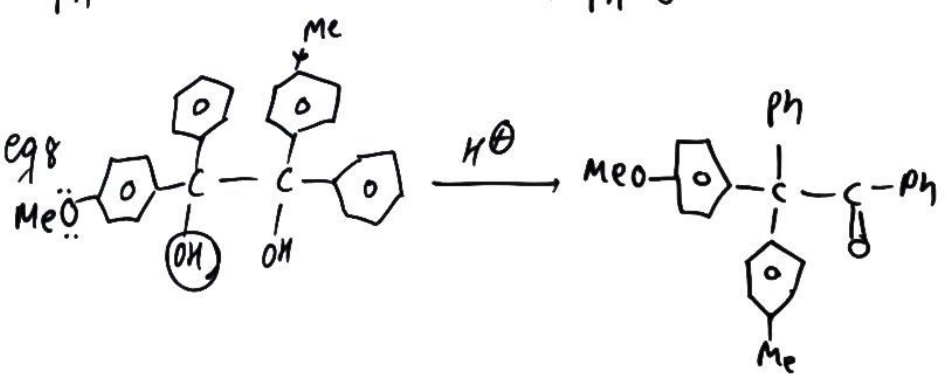
eg 6



eg 7



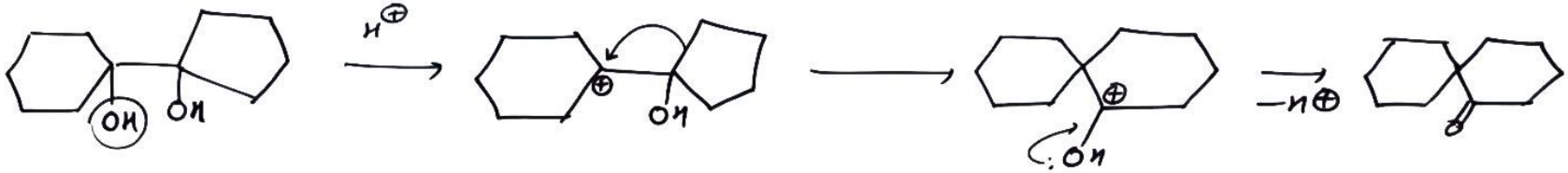
eg 8



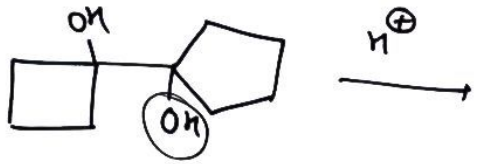
# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## Pinacol - Pinacolone Rearrangement:-

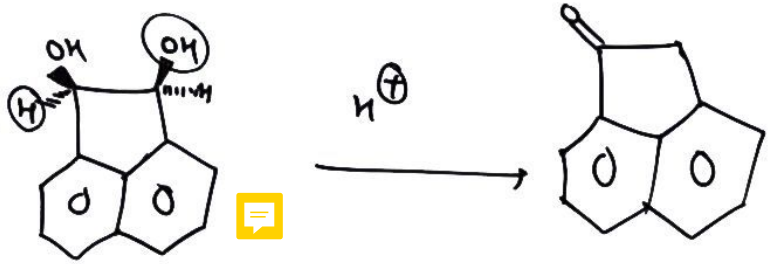
eg 10




eg 11



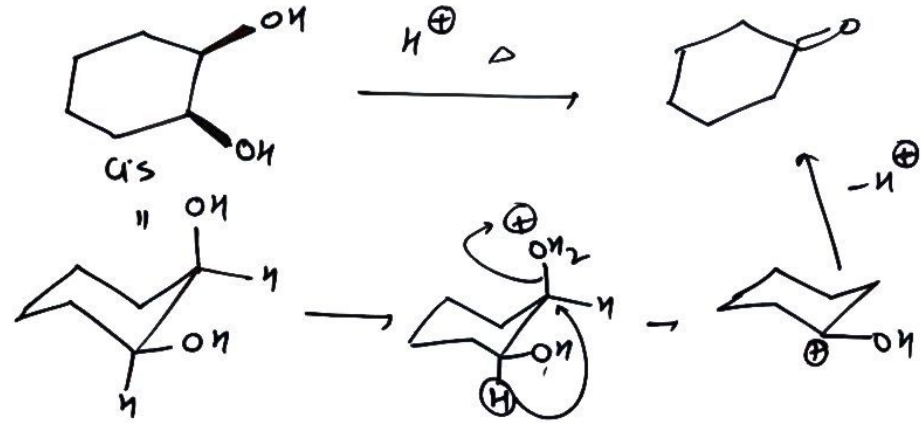

eg 12



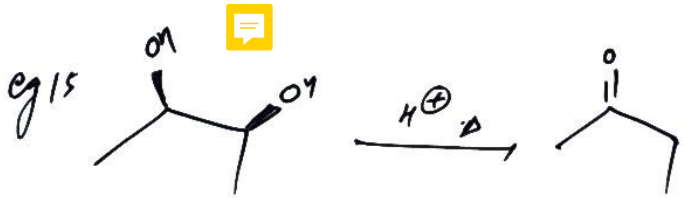
# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## Pinacol - Pinacolone Rearrangement:-

eg 13



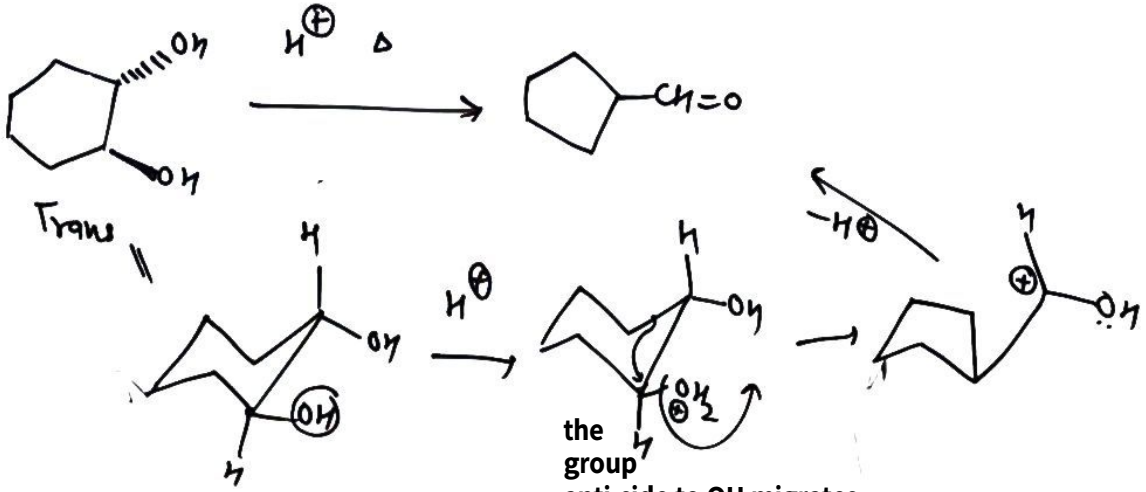

eg 15




eg 16




★ eg 14

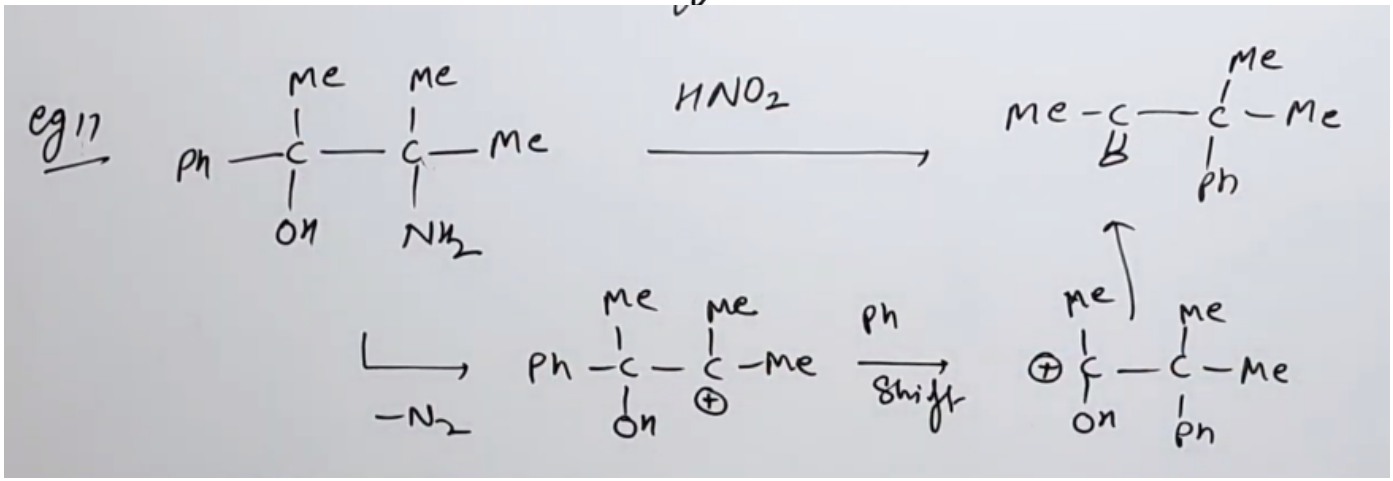



the group anti-side to OH migrates

# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## Semi Pinacol - Pinacolone Rearrangement:-

eg 17 →

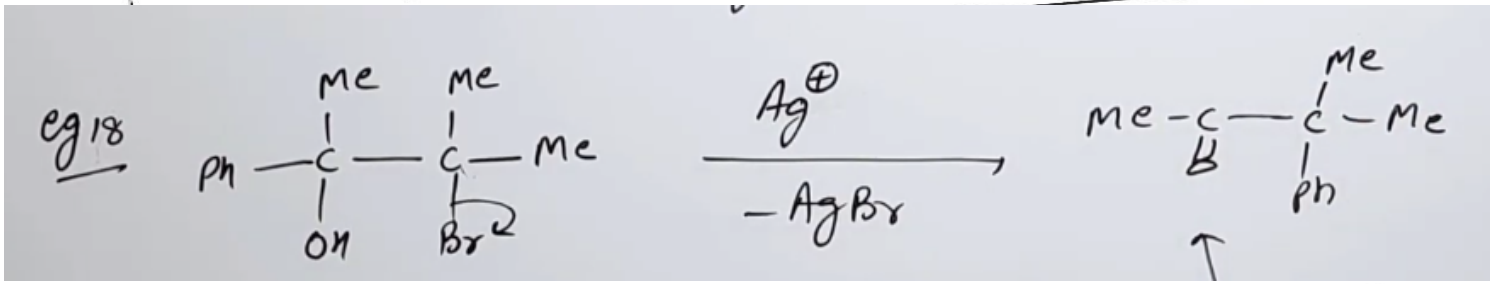



**Hint**

$$R-NH_2 \xrightarrow{HNO_2} R-\overset{\oplus}{N}\equiv N \longrightarrow R^{\oplus} + N_2$$

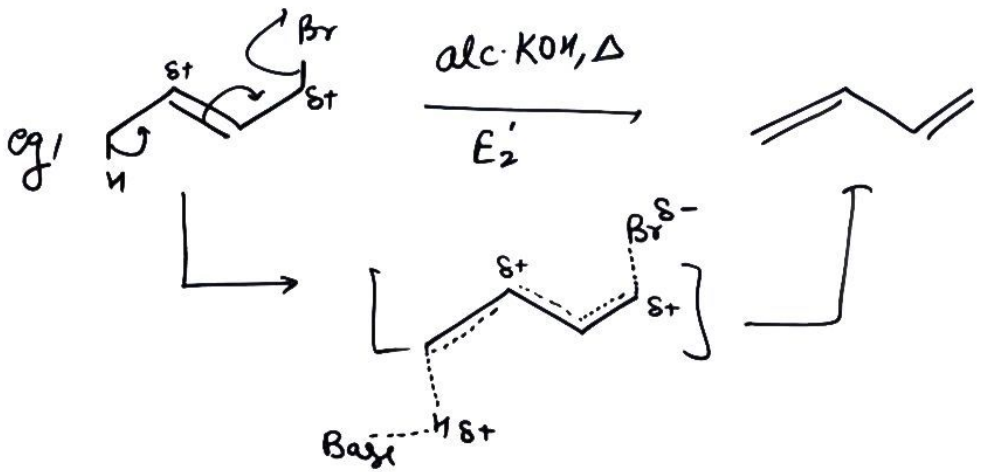
1° aliphatic amine — diazonium salt — i.g.

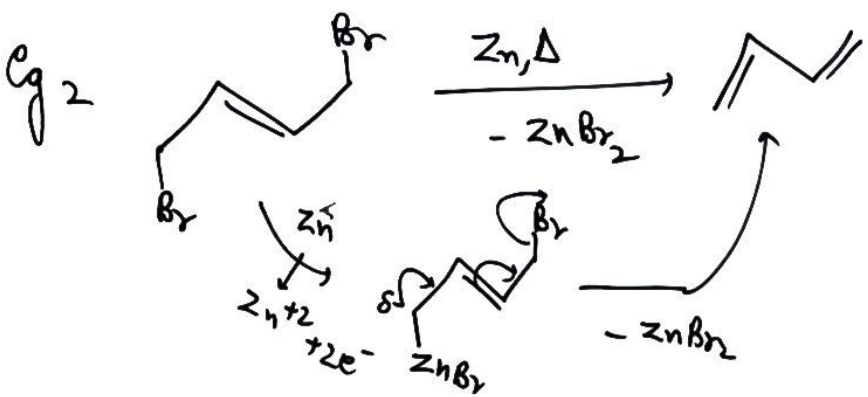
eg 18 →

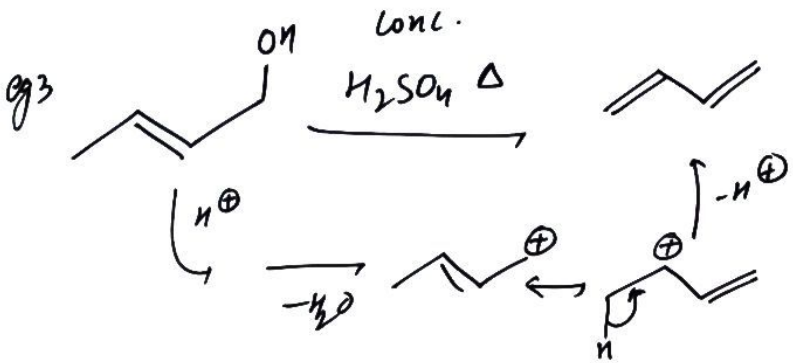


# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## (c) 1,4- Elimination = conjugate elimination:-



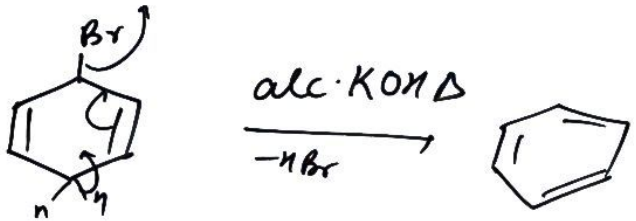



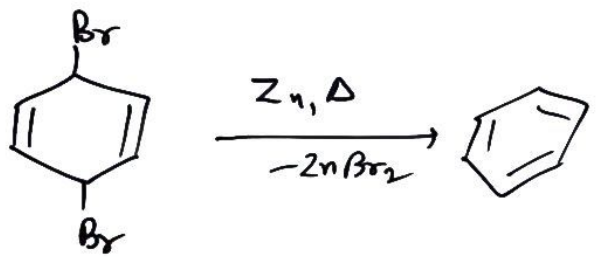
# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## (c) 1,4- Elimination = conjugate elimination:-

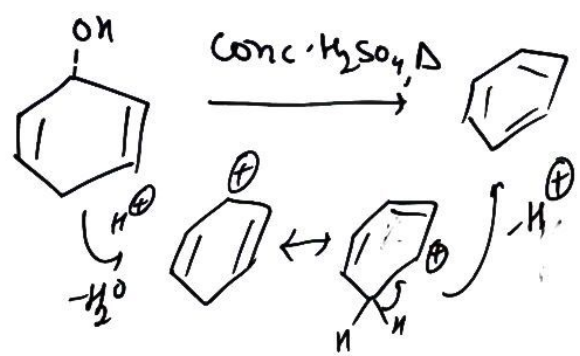
Eg 4



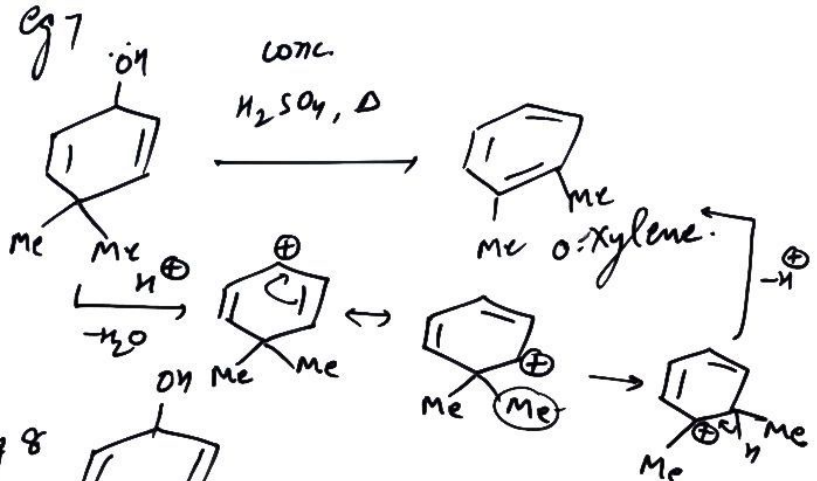

Eg 5



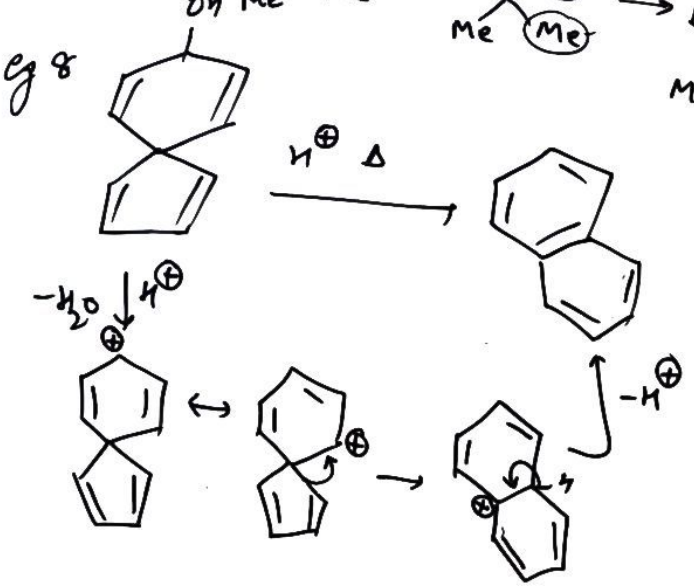

Eg 6




Eg 7




Eg 8



# ORGANIC REACTION MECHANISM - IV (ORM-IV)

## Dienone - Phenol Rearrangement

unrelated to elimination

